

# Notes on [Bam20a]: a brief account on $\epsilon$ -regularity theorems in Ricci flow

Kyeongho Bang

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Below are my notes on Richard Bamler's 2020 paper *Entropy and Heat Kernel Bounds on a Ricci Flow Background*. Most the results/proofs/computations are from the main references, presented in the notes possibly with slight modifications.\*

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\*Main references:

- Kleier-Lott, *Notes on Perelman's Papers*
- Hein-Naber, *New Logarithmic Sobolev Inequalities and an  $\epsilon$ -Regularity Theorem for the Ricci Flow*
- Richard Bamler, *Entropy and Heat Kernel Bounds on a Ricci Flow Background*

# 1 $\epsilon$ -regularity theorem in smooth Ricci flow

Bamler's paper [RB20a] is essentially an improvement of Hein-Naber's  $\epsilon$ -regularity theorem, originally introduced and proved in 2014. In this section, we trace the brief historical development of the  $\epsilon$ -regularity theorem in the smooth setting, beginning from Perelman's  $\mathcal{W}$ -entropy.

## 1.1 Perelman's $\mathcal{W}$ -functional and no local collapsing theorem

In this subsection, we introduce Perelman's  $\mathcal{W}$ -functional and a version of local collapsing theorem, whose proof utilizes the monotonicity of  $\mathcal{W}$ -functional. Main reference for this subsection is Kleiner-Lott's *Notes on Perelman's Paper*.

Notations (in this subsection):

- $M$ : closed manifold
- $\mathcal{M}$ : space of smooth metrics on  $M$  (formally an infinite-dimensional manifold)
- $T_g\mathcal{M}$  = set of symmetric covariant 2-tensors  $v_{ij}$  on  $M$
- $T_f C^\infty(M) = C^\infty(M)$ .
- $dV = d\text{vol}_g$ ,  $\Delta = \text{div}(\text{grad})$ .

### 1.1.1 $\mathcal{F}$ -functional and its monotonicity

Using the  $\mathcal{F}$ -functional, Perelman reformulated Ricci flow as a gradient flow on  $\mathcal{M}$ .

**Definition 1.1** ( $\mathcal{F}$ -functional). *The  $\mathcal{F}$ -functional  $\mathcal{F} : \mathcal{M} \times C^\infty(M) \rightarrow \mathbb{R}$  is defined as*

$$\mathcal{F}(g, f) := \int_M \left( R + |\nabla f|^2 \right) e^{-f} dV. \quad (1.1)$$

We derive the differential (variation) of  $\mathcal{F}$ . Given  $v_{ij} \in T_g\mathcal{M}$  and  $h \in C^\infty(M)$ , the differential of  $\mathcal{F}$  acting on  $v_{ij}$ ,  $h$  is written as  $\delta\mathcal{F}(v_{ij}, h)$ . Write  $v = g^{ij}v_{ij}$ .

**Proposition 1.2** (Differential of  $\mathcal{F}$ ).

$$\delta\mathcal{F}(v_{ij}, h) = \int_M e^{-f} \left[ -v_{ij}(R_{ij} + \nabla_i \nabla_j f) + \left( \frac{v}{2} - h \right) (2\Delta f - |\nabla f|^2 + R) \right] dV \quad (1.2)$$

*Proof.* Proof follows from direct computation. For the sake of completeness, below we derive the variations of curvature terms. Given a smooth metric  $g = g_{ij}$ , a symmetric covariant 2-tensor  $V = v_{ij}$ , for a sufficiently small  $t > 0$ , we have

$$(g + tV)^{-1} = g^{-1} \left( I + \sum_{k \geq 1} (-1)^k t^k (gV)^k \right) = g^{-1} - tV + o(t^2). \quad (1.3)$$

From this see that  $\delta g(V) = V$  and  $\delta g^{-1}(V) = -V$ . Now we compute the variation of Christoffel

symbol  $\tilde{\Gamma}_{ij}^k(t)$  of  $\tilde{g}_t = g + tV$ :

$$\left. \frac{d}{dt} \right|_{t=0} \tilde{\Gamma}_{ij}^k(t) = -\frac{1}{2}v_{kl} \left( \frac{\partial g_{il}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right) + \frac{1}{2}g^{kl} \left( \frac{\partial v_{il}}{\partial x^j} + \frac{\partial v_{jl}}{\partial x^i} - \frac{\partial v_{ij}}{\partial x^l} \right) \quad (1.4)$$

$$= \delta\Gamma_{ij}^k(V) \quad (1.5)$$

$$= \frac{1}{2} \left( \frac{\partial v_{ik}}{\partial x^j} + \frac{\partial v_{jk}}{\partial x^i} - \frac{\partial v_{ij}}{\partial x^k} \right) \quad (\text{in normal coordinates } x^i \text{ at a point}) \quad (1.6)$$

Riemannian tensor:

$$\delta R_{ijk}^l = \partial_i \delta \Gamma_{kj}^l - \partial_j \delta \Gamma_{ki}^l - \delta \Gamma_{pj}^l \cdot \Gamma_{ki}^p - \Gamma_{pj}^l \cdot \delta \Gamma_{ki}^p + \delta \Gamma_{pi}^l \cdot \Gamma_{kj}^p + \Gamma_{pi}^l \cdot \delta \Gamma_{kj}^p \quad (1.7)$$

$$(\text{at } V) = \partial_i \left( \frac{\partial v_{jl}}{\partial x^k} - \frac{\partial v_{kj}}{\partial x^l} \right) - \partial_j \left( \frac{\partial v_{il}}{\partial x^k} - \frac{\partial v_{ki}}{\partial x^l} \right) \quad \text{in normal coordinates.} \quad (1.8)$$

Ricci tensor:

$$\delta R_{ij} = \delta (g^{kl} R_{kijl}) \quad (1.9)$$

$$= \delta (g^{kl} g_{lp} R_{kij}^p) \quad (1.10)$$

$$= \delta g^{kl} \cdot g_{lp} \cdot R_{kij}^p + g^{kl} \cdot \delta g_{lp} \cdot R_{kij}^p + \delta R_{kij}^k \quad (1.11)$$

$$(\text{at } V) = -v_{kl} g_{lp} R_{kij}^p + g^{kl} v_{lp} R_{kij}^p + \delta R_{kij}^k \quad (1.12)$$

$$(\text{in normal coordinates}) = -v_{kl} R_{kij}^l + v_{lk} R_{kji}^l + \delta R_{kij}^k \quad (1.13)$$

$$= \delta R_{kij}^k = \partial_k \left( \frac{\partial v_{ik}}{\partial x^j} - \frac{\partial v_{ij}}{\partial x^k} \right) - \partial_i \left( \frac{\partial v_{kk}}{\partial x^j} - \frac{\partial v_{jk}}{\partial x^k} \right). \quad (1.14)$$

Scalar curvature:

$$\delta R = \delta (g^{ij} R_{ij}) \quad (1.15)$$

$$= -v_{ij} R_{ij} + g^{ij} \delta R_{ij}. \quad (1.16)$$

Note that in normal coordinates,

$$g^{ij} \delta R_{ij} = \sum_i \delta R_{ii} = 2 \sum_{i,k} (\partial_i \partial_k v_{ik} - \partial_i \partial_i v_{kk}).$$

From the above computation, we deduce

$$\delta R = -\Delta v + \nabla_i \nabla_j v_{ij} - R_{ij} v_{ij} \quad (1.17)$$

$$\delta |\nabla f|^2 = -v^{ij} \nabla_i f \nabla_j f + 2 \langle \nabla f, \nabla h \rangle \quad (1.18)$$

$$\delta(e^{-f} dV) = \left( \frac{v}{2} - h \right) e^{-f} dV. \quad (1.19)$$

Carrying out integration by parts and substituting (1.17)-(1.19), we obtain the desired formula.  $\square$

From (1.19) we notice that, if we can fix  $e^{-f} dV$ , then we can get rid of the term

$$\left( \frac{v}{2} - h \right) (2\Delta f - |\nabla f|^2 + R)$$

in the equation (1.2). Define a section  $s : \mathcal{M} \rightarrow \mathcal{M} \times C^\infty(M)$ ,  $s(g) = (g, \ln(\frac{dV}{dm}))$ , where  $dm = e^{-f}dV$  is some fixed smooth measure on  $M$  ( $dV = dV_g$ ). Then  $\mathcal{F}^m := \mathcal{F} \circ s$  is a functional on  $\mathcal{M}$  and

$$\delta \mathcal{F}^m(v_{ij}) = \int_M -v_{ij} (R_{ij} + \nabla_i \nabla_j f) e^{-f} dV. \quad (1.20)$$

We are ready to give a gradient flow formulation of Ricci flow. Define a formal Riemannian metric on  $\mathcal{M}$  by

$$\langle v_{ij}, v_{ij} \rangle_g = \frac{1}{2} \int_M v^{ij} v_{ij} dm. \quad (1.21)$$

The **gradient flow of  $\mathcal{F}^m$  on  $\mathcal{M}$**  is given by

$$(g_{ij})_t = -2(R_{ij} + \nabla_i \nabla_j f). \quad (1.22)$$

The **induced flow equation** for  $f$  is

$$f_t = \frac{\partial}{\partial t} \ln \left( \frac{dV}{dm} \right) = \frac{1}{2} g^{ij} (g_{ij})_t = -\Delta f - R. \quad (1.23)$$

It follows that

$$\frac{d}{dt} \mathcal{F}^m = \mathcal{F}_t^m = 2 \int_M |R_{ij} + \nabla_i \nabla_j f|^2 dm. \quad (1.24)$$

Now we wish to transform (1.22) into the Ricci flow equation via a series of time-dependent diffeomorphisms. Let  $V(t)$  be the time-dependent vector field which generates the desired diffeomorphisms:

$$\begin{cases} (g_{ij})_t = -2(R_{ij} + \nabla_i \nabla_j f) + \mathcal{L}_V g \\ f_t = -\Delta f - R + \mathcal{L}_V f \end{cases} \quad (1.25)$$

where  $g_t = g \circ \Phi_t$ ,  $f(t) = f \circ \Phi_t$ ,  $\Phi_t$  is the diffeomorphism generated by  $V(t)$ . Taking  $V = \nabla f$ :

$$\begin{cases} (g_{ij})_t = -2R_{ij} \\ f_t = -\Delta f - R + |\nabla f|^2. \end{cases} \quad (1.26)$$

Since  $\mathcal{F}(g, f)$  is unchanged by a simultaneous pullback of  $g$ ,  $f$ , and so for (1.24), it follows that under the evolution equation (1.26) we have

**Proposition 1.3** (Monotonicity of  $\mathcal{F}$ -functional).

$$\frac{d}{dt} \mathcal{F}(g(t), f(t)) = 2 \int_M |R_{ij} + \nabla_i \nabla_j f|^2 e^{-f} dV. \quad (1.27)$$

Note that  $g$  and  $f$  are no longer related by  $e^{-f}dV = dm$ .

**Remark 1.4.** The second equation in (1.26) can be rewritten as a backward heat equation

$$\partial_t e^{-f(t)} = -\Delta e^{-f(t)} + R e^{-f}. \quad (1.28)$$

This equation can be solved from the solution of the first equation on an interval  $[t_1, t_2]$ .

### 1.1.2 $\lambda$ -functional and Ricci flow as a gradient flow

We introduce the functional  $\lambda(\cdot)$  and show that  $\lambda$  is a monotonic quantity under the Ricci flow. This will lead up to showing that a steady breather on a compact manifold is a gradient Ricci folion.

We begin with the following important observation.

**Proposition 1.5.** *Given a metric  $g$ ,  $\exists$  a unique minimizer  $\bar{f}$  of  $\mathcal{F}(g, f)$  under the constraint*

$$\int_M e^{-f} dV = 1. \quad (1.29)$$

*Proof.* Write  $\mathcal{F}(g, f) = \int_M (Re^{-f} + 4|\nabla e^{-f/2}|^2) dV$ . Putting  $\Phi = e^{-f/2}$ ,

$$\mathcal{F}(g, f) = \int_M \Phi (-4\Delta\Phi + R\Phi) dV \quad (\text{integration by parts.}) \quad (1.30)$$

Then we see that the constraint equation is equivalent to  $\int_M \Phi^2 dV = 1$ . Let  $\lambda$  denote the smallest eigenvalue of the operator  $(-4\Delta + R)$  and  $e^{\bar{f}/2}$  a normalized eigenvector of the operator. Since the operator is a Schrödinger-type operator, it always has a unique minimizer.  $\square$

We define the  $\lambda$ -functional as in the above proof.

**Definition 1.6** ( $\lambda$ -functional). *The  $\lambda$ -functional is given by  $\lambda(g) = \mathcal{F}(g, \bar{f})$ .*

**Remark 1.7.** *From the eigenvalue perturbation theory, we know that if  $g(t)$  is a smooth family of metrics, then  $\lambda(g(t))$  and  $\bar{f}(t)$  are smooth in  $t$ .*

Similar to the  $\mathcal{F}$ -functional,  $\lambda$ -functional also satisfies following monotonicity:

**Proposition 1.8** (Monotonicity of  $\lambda$ -functional). *If  $g(\cdot)$  is a solution to the Ricci flow, then  $\lambda(g(t))$  is non-decreasing in  $t$ .*

*Proof.* Consider a time interval  $[t_1, t_2]$  and the minimizer  $\bar{f}(t_2)$ . In particular,  $\lambda(t_2) = \mathcal{F}(g(t_2), \bar{f}(t_2))$ . Put  $u(t_2) = e^{-\bar{f}(t_2)}$ . Then we can solve the backward heat equation

$$\partial_t u = -\Delta u + Ru$$

backwards in the interval  $[t_1, t_2]$  to obtain a solution  $u(\cdot, t)$ . Once we have shown  $u(\cdot, t') > 0$  for all  $t' \in [t_1, t_2]$ , we capture the monotonicity of  $\lambda$ : for  $t \in [t_1, t_2]$  define  $f(t)$  by  $u(t) = e^{-f(t)}$ . From the definition of  $u$ , we know that  $e^{-f(t_2)} = e^{-\bar{f}(t_2)}$ . And from (1.27) (the monotonicity of  $\mathcal{F}$ ), we know that

$$\mathcal{F}(g(t_1), f(t_1)) \leq \mathcal{F}(g(t_2), f(t_2)) = \mathcal{F}(g(t_2), \bar{f}(t_2)) = \lambda(t_2). \quad (1.31)$$

By the definition of  $\lambda$ ,  $\lambda(t_1) = \mathcal{F}(g(t_1), \bar{f}(t_1)) \leq \mathcal{F}(g(t_1), f(t_1))$ , which implies  $\lambda(t_1) \leq \lambda(t_2)$ . Now we are left to prove  $u(\cdot, t') > 0$ .

Take any  $t' \in [t_1, t_2)$ ,  $x' \in M$ . Let  $h$  denote the solution to the forward heat equation  $\partial_t h = \Delta h$  on  $(t', t_2]$  with  $\lim_{t \rightarrow t'} h(t) = \delta_{x'}$ . We have

$$\frac{d}{dt} \int_M u(\cdot, t) h(\cdot, t) dV = \int_M (\partial_t u + \Delta u - Ru) h + u(\partial_t h - \Delta h) dV = 0 \quad (1.32)$$

where the term  $\Delta u - Ru$  comes from the variation of  $dV_{g(t)}$ . Since  $h(\cdot, t) > 0$  for all  $t' < t \leq t_2$ :

$$u(x', t') = \int_M u(x, t') \delta_{x'}(x) dV(x) = \lim_{t \rightarrow t'} \int_M u(\cdot, t) h(\cdot, t) dV \equiv \int_M u(\cdot, t_2) h(\cdot, t_2) dV > 0 \quad (1.33)$$

where the second to last  $\equiv$  comes from (1.32). This completes our proof.  $\square$

In fact, we can obtain a sharper estimate of **Proposition 1.8**.

**Lemma 1.9.**

$$\frac{d\lambda}{dt} \geq \frac{2}{n} \lambda^2(t).$$

*Proof.* Given a time interval  $[t_1, t_2]$ , with the same notation as the proof of **Proposition 1.8**, we have

$$\lambda(t_1) \leq \mathcal{F}(g(t_1), f(t_1)) = \mathcal{F}(g(t_2), f(t_2)) - 2 \int_{t_1}^{t_2} \int_M |R_{ij} + \nabla_i \nabla_j f|^2 e^{-f} dV dt \quad (1.34)$$

$$= \lambda(t_2) - 2 \int_{t_1}^{t_2} \int_M |R_{ij} + \nabla_i \nabla_j f|^2 e^{-f} dV dt \quad (1.35)$$

as  $e^{-f}$  solves the backward heat equation with final value  $e^{-f(t_2)} = e^{-\bar{f}(t_2)}$ . This implies

$$\left. \frac{d\lambda}{dt} \right|_{t=t_2} = \lim_{t_1 \rightarrow t_2^-} \frac{\lambda(t_2) - \lambda(t_1)}{t_2 - t_1} \geq 2 \int_M |R_{ij} + \nabla_i \nabla_j f|^2 e^{-f} dV \Big|_{t=t_2} \quad (1.36)$$

$$= 2 \int_M |R_{ij} + \nabla_i \nabla_j f|^2 e^{-\bar{f}} dV \quad (1.37)$$

$$\stackrel{\text{trace}}{\geq} \frac{2}{n} \int_M (R + \Delta \bar{f})^2 e^{-\bar{f}} dV. \quad (1.38)$$

By Cauchy-Schwarz inequality and the constraint  $\int_M e^{-\bar{f}} = 1$ , we have

$$\left( \int_M (R + \Delta \bar{f})^2 e^{-\bar{f}} dV \right)^2 \leq \int_M (R + \Delta \bar{f})^2 e^{-\bar{f}} dV. \quad (1.39)$$

Finally, with integration by parts,

$$\int_M (R + \Delta \bar{f}) e^{-\bar{f}} dV = \int_M (R + \Delta \bar{f}) e^{-\bar{f}} dV = \lambda(t). \quad (1.40)$$

$\square$

Now we present the rescaled  $\lambda$ -functional,  $\bar{\lambda}(g) := \lambda(g) V(g)^{2/n}$ , which is scale-invariant, therefore is constant in  $t$  along steady/shrinking/expanding solitons.

**Proposition 1.10** (Monotonicity of  $\bar{\lambda}$ -functional). *If  $g(\cdot)$  is a solution to Ricci flow and  $\bar{\lambda}(g(t)) \leq 0$  for some  $t$ , then*

$$\frac{d}{dt} \bar{\lambda}(g(t)) \geq 0. \quad (1.41)$$

*Proof.* Proof follows from direct computation.

We have

$$\frac{d\bar{\lambda}}{dt} = \frac{d\lambda}{dt} V(t)^{2/n} - \frac{2}{n} V(t)^{\frac{2-n}{n}} \lambda(t) \int_M R dV. \quad (1.42)$$

From (1.36),

$$\frac{d\bar{\lambda}}{dt} \geq V(t)^{2/n} \left[ 2 \int_M |R_{ij} + \nabla_i \nabla_j \bar{f}|^2 e^{-\bar{f}} dV - \frac{2}{n} V(t)^{-1} \lambda(t) \int_M R dV \right]. \quad (1.43)$$

Since the function  $\ln V(t)$  is spatially constant, using it as a test function for  $\mathcal{F}$  yields

$$\lambda(t) \leq V(t)^{-1} \int_M R dV = \mathcal{F}(g, \ln V(t)). \quad (1.44)$$

As we have assumed  $\lambda(t) \leq 0$ :

$$-\lambda(t)^2 \leq -V(t)^{-1} \lambda(t) \int_M R dV \quad (1.45)$$

which implies

$$\frac{d\bar{\lambda}}{dt} \geq V(t)^{2/n} \left[ 2 \int_M |R_{ij} + \nabla_i \nabla_j \bar{f}|^2 e^{-\bar{f}} dV - \frac{2}{n} \lambda(t)^2 \right]. \quad (1.46)$$

Since

$$|R_{ij} + \nabla_i \nabla_j \bar{f}|^2 \geq \left| R_{ij} + \nabla_i \nabla_j \bar{f} - \frac{1}{n} (R + \Delta \bar{f}) g_{ij} \right|^2 + \frac{1}{n} (R + \Delta \bar{f})^2, \quad (1.47)$$

using (1.40), we obtain

$$\begin{aligned} \frac{d\bar{\lambda}}{dt} &\geq 2V(t)^{2/n} \left[ \int_M \left| R_{ij} + \nabla_i \nabla_j \bar{f} - \frac{1}{n} (R + \Delta \bar{f}) g_{ij} \right|^2 e^{-\bar{f}} dV \right. \\ &\quad \left. + \frac{1}{n} \int_M (R + \Delta \bar{f})^2 e^{-\bar{f}} dV - \frac{1}{n} \left( \int_M (R + \Delta \bar{f}) e^{-\bar{f}} dV \right)^2 \right]. \end{aligned} \quad (1.48)$$

Applying Cauchy-Schwarz and normalization property, we deduce that the RHS term of (1.48) is nonnegative.  $\square$

In 1.1.1, we have presented the gradient flow formulation of Ricci flow using the  $\mathcal{F}^m$ -functional. Here we give another gradient flow formulation of Ricci flow using  $\lambda$ -functional.

We can consider  $\lambda$ -functional as a function on  $\mathcal{M}$ . Define a formal metric on  $\mathcal{M}$ :  $\forall v_{ij} \in T_g \mathcal{M}$ ,

$$\langle v_{ij}, v_{ij} \rangle := \frac{1}{2} \int_M v^{ij} v_{ij} \Phi^2 dV(g) \quad (1.49)$$

where  $\Phi = \Phi(g)$  is the unique normalized positive eigenvector corresponding to  $\lambda(g)$ .

**Lemma 1.11** (Gradient flow of  $\lambda$ ). *The formal gradient flow of  $\lambda$  is*

$$\frac{\partial g_{ij}}{\partial t} = -2 (R_{ij} - \nabla_i \nabla_j \ln \Phi). \quad (1.50)$$

*Proof.* Set

$$\lambda(g) = \inf_{f \in C^\infty(M), \int e^{-f} = 1} \mathcal{F}(g, f). \quad (1.51)$$

Consider the variations  $\delta g_{ij} = v_{ij}$ ,  $\delta f = h$ . Let  $f$  be the minimizer in (1.51). Then

$$0 = \delta \left( \int_M e^{-f} dV \right) = \int_M \left( \frac{v}{2} - h \right) e^{-f} dV. \quad (1.52)$$

Using (1.2) (the variation of  $\mathcal{F}$ -functional), we can derive the variation of  $\lambda$ -functional:

$$\delta \lambda(v_{ij}) = \int_M e^{-f} \left[ -v_{ij}(R_{ij} + \nabla_i \nabla_j f) + \left( \frac{v}{2} - h \right) (2\Delta f - |\nabla f|^2 + R) \right] dV. \quad (1.53)$$

As  $\Phi = e^{-f/2}$  satisfies

$$-4\Delta \Phi + R\Phi = \lambda\Phi \quad (1.54)$$

it follows that

$$2\Delta f - |\nabla f|^2 + R = \lambda \quad (1.55)$$

which implies

$$\delta \lambda(v_{ij}) = - \int_M e^{-f} v_{ij} (R_{ij} + \nabla_i \nabla_j f) dV. \quad (1.56)$$

Hence the corresponding gradient flow is

$$\partial_t g_{ij} = -2(R_{ij} + \nabla_i \nabla_j f). \quad (1.57)$$

□

### 1.1.3 $\mathcal{W}$ -functional and no local collapsing theorem

Now we introduce the  $\mathcal{W}$ -functional, also known as the  $\mathcal{W}$ -entropy functional, which is an essential ingredient for proving “no local collapsing theorem”.

**Definition 1.12** ( $\mathcal{W}$ -functional). *The  $\mathcal{W}$ -functional  $\mathcal{W} : \mathcal{M} \times C^\infty(M) \times \mathbb{R}^+ \rightarrow \mathbb{R}$  is given by*

$$\mathcal{W}(g, f, \tau) := \frac{1}{(4\pi\tau)^{n/2}} \int_M \left[ \tau(|\nabla f|^2 + R) + f - n \right] e^{-f} dV. \quad (1.58)$$

Different from other functionals,  $\mathcal{W}$ -functional has certain type of symmetry:  $\forall \varphi \in \text{Diff}(M)$ ,  $\mathcal{W}(\varphi^* g, \varphi^* f, \tau) = \mathcal{W}(g, f, \tau)$  and  $\mathcal{W}(cg, f, c\tau) = \mathcal{W}(g, f, \tau)$  for all  $c > 0$ .

Like other functionals,  $\mathcal{W}$ -functional also satisfies monotonicity. To do this, we first derive the variaton of  $\mathcal{W}$ .

**Proposition 1.13** (Variation of  $\mathcal{W}$ ). *Write  $\delta g_{ij} = v_{ij}$ ,  $\delta f = h$ ,  $\delta \tau = \sigma$ .*

$$\begin{aligned} \delta \mathcal{W}(v_{ij}, h, \sigma) = & \frac{1}{(4\pi\tau)^{n/2}} \int_M \left[ \sigma(R + |\nabla f|^2) - \tau v_{ij}(R_{ij} + \nabla_i \nabla_j f) + h \right. \\ & \left. + \left( \tau (2\Delta f - |\nabla f|^2 + R) + f - n \right) \cdot \left( \frac{v}{2} - h - \frac{n\sigma}{2\tau} \right) \right] e^{-f} dV. \end{aligned} \quad (1.59)$$

*Proof.* Direct computation. □

Now we fix a smooth measure  $dm$  on  $M$  with mass 1 and relate  $f$  to  $g$  and  $\tau$  by requiring

$$(4\pi\tau)^{-n/2}e^{-f}dV = dm. \quad (1.60)$$

Then  $\frac{v}{2} - h - \frac{n\sigma}{2\tau} = 0$  and

$$\delta\mathcal{W} = (4\pi\tau)^{-\frac{n}{2}} \int_M \left[ \sigma(R + |\nabla f|^2) - \tau v_{ij}(R_{ij} + \nabla_i \nabla_j f) + h \right] e^{-f} dV. \quad (1.61)$$

Also, we consider  $\frac{d\mathcal{W}}{dt}$  when

$$\begin{cases} (g_{ij})_t = -2(R_{ij} + \nabla_i \nabla_j f) \\ f_t = -\Delta f - R + \frac{n}{2\tau} \\ \tau_t = -1, \end{cases} \quad (1.62)$$

i.e. treating  $f, g, \tau$  as functions of  $t \in \mathbb{R}_{\geq 0}$  flowing by the above equations.

To apply (1.61), we put

$$\begin{cases} v_{ij} = -2(R_{ij} + \nabla_i \nabla_j f) \\ h = -\Delta f - R + \frac{n}{2\tau} \\ \sigma = -1. \end{cases} \quad (1.63)$$

$$\implies \frac{d\mathcal{W}}{dt} = (4\pi\tau)^{-\frac{n}{2}} \int_M 2\tau \left| R_{ij} + \nabla_i \nabla_j f - \frac{1}{2\tau} g_{ij} \right|^2 e^{-f} dV. \quad (1.64)$$

Same as what we have done for the  $\mathcal{F}$ -functional, adding a Lie derivative to (1.62), we obtain the new flow equations

$$\begin{cases} (g_{ij})_t = -2R_{ij} \\ f_t = -\Delta f + |\nabla f|^2 - R + \frac{n}{2\tau} \\ \tau_t = -1 \end{cases} \quad (1.65)$$

with (1.64) still holding. Note that while we no longer have the relationship  $(4\pi\tau)^{-n/2}e^{-f}dV = dm$  for all  $t$  (having the diffeomorphism induced by  $\nabla f$  acted on  $g$ ), we do have

$$\int_M (4\pi\tau)^{-n/2}e^{-f}dV \equiv 1, \quad (1.66)$$

which is what all we needed.

Now we can consider the variational problem: minimization of  $\mathcal{W}(g, f, \tau)$  under the constraint  $\int_M (4\pi\tau)^{-n/2}e^{-f}dV = 1$ . We write

$$\mu(g, \tau) := \inf_f \left\{ \mathcal{W}(g, f, \tau) : \int_M (4\pi\tau)^{-n/2}e^{-f}dV = 1 \right\}. \quad (1.67)$$

After a change of variables  $\Phi = e^{-f/2}$ , it is equivalent to minimizing

$$(4\pi\tau)^{-n/2} \int_M \left[ \tau \left( 4|\nabla \Phi|^2 + R\Phi^2 \right) - 2\Phi^2 \log \Phi - n\Phi^2 \right] dV \quad (1.68)$$

under the constraint  $(4\pi\tau)^{-n/2} \int_M \Phi^2 dV = 1$ . By a standard theory of log-Sobolev type equations (ref. O. Rothaus 80', *Section 1*), the infimum of (1.68)  $\Phi$  exists, is finite and positive. Notice that such minimizer  $\Phi$  will be a weak solution of the variational equation

$$\tau(-4\Delta + R)\Phi = 2\Phi \log \Phi + (\mu(g, \tau) + n)\Phi. \quad (1.69)$$

From the elliptic regularity theory, we know that  $\Phi$  is smooth, therefore  $f = -2 \log \Phi$  is also smooth.

Following the argument in the case of  $\lambda$ -functional, we also deduce that

**Proposition 1.14.**  *$\mu(g(t), t_0 - t)$  is nondecreasing in  $t < t_0$  for a Ricci flow solution.*

*Proof.* Let  $[t_1, t_2]$  be a time interval where our Ricci flow is defined on, where  $t_2 < t_0$ . Consider the minimizer  $\bar{f}(t_2)$  of  $\mathcal{W}$ :

$$\mathcal{W}(g(t_2), \bar{f}(t_2), t_0 - t_2) = \mu(g(t_2), t_0 - t_2).$$

Put  $u(t_2) = e^{-\bar{f}(t_2)}$ , and solve the backward heat equation

$$\partial_t u = -\Delta u + Ru \text{ on } [t_1, t_2].$$

Let  $x' \in M$ ,  $t' \in [t_1, t_2]$ , and  $h$  to be the solution to the forward heat equation

$$\partial_t h = \Delta h$$

on  $(t', t_2]$  with  $\lim_{t \rightarrow t'} h(t) = \delta_{x'}$ . Then

$$\frac{d}{dt} \int_M u(t) h(t) dV_{g(t)} = \int_M (\partial_t u + \Delta u - Ru) h + u (\partial_t h - \Delta h) dV_{g(t)} = 0.$$

Since  $h(t) > 0$  for  $t \in (t', t_2]$ , we have

$$u(x', t') = \int_M u(x, t') \delta_{x'}(x) dV_{g(t')}(x) = \lim_{t \rightarrow t'} \int_M u(t) h(t) dV_{g(t)} = \int_M u(t_2) h(t_2) dV_{g(t_2)} > 0.$$

For  $t \in [t_1, t_2]$ , define  $f(t)$  by  $u(t) = e^{-f(t)}$ . Due to (2.5), we have

$$\mathcal{W}(g(t_1), f(t_1), t_0 - t_1) \leq \mathcal{W}(g(t_2), f(t_2), t_0 - t_2) = \mu(g(t_2), t_0 - t_2).$$

Then it follows from the definition of  $\mu$  that

$$\mu(g(t_1), t_0 - t_1) \leq \mathcal{W}(g(t_1), f(t_1), t_0 - t_1) \leq \mu(g(t_2), t_0 - t_2),$$

which is what we wanted. □

Now that we have established the monotonicity of  $\mu$ , we are ready to prove (a version of) no local collapsing theorem.

**Definition 1.15** (Locally collapsing). *A smooth Ricci flow solution  $g(\cdot)$  on a time interval  $[0, T)$ ,  $0 < T \leq \infty$  is said to be **locally collapsing at  $T$**  if there exists a sequence of times  $t_k \rightarrow T$  and a*

sequence of metric balls  $B_k = B(p_k, r_k)$  at times  $t_k$  such that

1.  $r_k^2/t_k$  is bounded,
2.  $|Rm|(g(t_k)) \leq r_k^{-2}$  in  $B_k$ , and
3.  $\lim_{k \rightarrow \infty} r_k^{-n} \text{Vol}(B_k) = 0$ .

**Theorem 1.16** (No local collapsing I). *If  $M$  is closed and  $T < \infty$ , then  $g(\cdot)$  is not locally collapsing at  $T$ .*

*Proof.* We argue by contradiction. Suppose  $g$  locally collapses at  $T < \infty$ .

Write  $\Phi = e^{-f/2}$ , then

$$\mathcal{W}(g, f, \tau) = (4\pi\tau)^{-\frac{n}{2}} \int_M \left[ 4\tau |\nabla \Phi|^2 + (\tau R - 2 \ln \Phi - n) \Phi^2 \right] dV. \quad (1.70)$$

Take  $\Phi_k(x) = e^{-c_k/2} \phi(d_{t_k}(x, p_k)/r_k)$  where  $\phi : [0, \infty) \rightarrow [0, 1]$  is a nonincreasing function such that

$$\phi(s) = \begin{cases} = 1, & 0 \leq s \leq \frac{1}{2} \\ = 0, & s \geq 1 \end{cases}$$

and  $|\phi'(s)| \leq 10$  for  $s \in [\frac{1}{2}, 1]$ . Then  $\Phi_k(x)$  is a priori merely Lipschitz, but by a slight smoothing, we can incorporate  $\Phi_k$  in the variational formula to bound  $\mathcal{W}$  from above.

Define the normalization constants  $c_k$ :

$$e^{c_k} = \int_M (4\pi r_k^2)^{-n/2} \phi^2(d_{t_k}(x, p_k)/r_k) dV \leq (4\pi r_k^2)^{-n/2} \text{vol}(B_k). \quad (1.71)$$

Thus  $c_k \rightarrow -\infty$  when we assume  $g(\cdot)$  locally collapses at  $T$ . Next,

$$\mathcal{W}(g(t_k), f_k, r_k^2) = (4\pi r_k^2)^{-\frac{n}{2}} \int_M \left[ 4r_k^2 |\nabla \Phi_k|^2 + (r_k^2 R - 2 \ln \Phi_k - n) \Phi_k^2 \right] dV. \quad (1.72)$$

Denote  $A_k(s)$  to be the mass of the distance sphere  $S(p_k, r_k s)$  around  $p_k$ . Write

$$\bar{R}_k(s) = r_k^2 A_k(s)^{-1} \int_{S(p_k, r_k s)} R d\text{Area}. \quad (1.73)$$

Computing the integral (1.72) radially:

$$\mathcal{W}(g(t_k), f_k, r_k^2) = \frac{\int_0^1 [4(\phi'(s))^2 + (\bar{R}_k(s) + c_k - 2 \ln \phi(s) - n) \phi^2(s)] A_k(s) ds}{\int_0^1 \phi^2(s) A_k(s) ds} \quad (1.74)$$

In the way it is defined,  $4\phi'(s)^2 - 2 \ln \phi(s) \cdot \phi(s)$  vanishes if  $s \notin [\frac{1}{2}, 1]$ , and is bounded above by  $400 + \frac{1}{2} < 401$  if  $s \in [\frac{1}{2}, 1]$ . Note that the local collapsing condition induces curvature bounds  $\text{Ric}(g(t_k)) \geq -(n-1)r_k^{-2}$  and  $|Rm(g(t_k))| \leq r_k^{-2}$  in  $B_k$ . From the Bishop-Gromov inequality, we have

$$\begin{aligned}
\frac{\int_0^1 [4(\phi'(s))^2 - 2 \ln \phi(s) \cdot \phi^2(s)] A_k(s) ds}{\int_0^1 \phi^2(s) A_k(s) ds} &\leq 401 \frac{\text{vol}(B(p_k, r_k)) - \text{vol}(B(p_k, r_k/2))}{\text{vol}(B(p_k, r_k/2))} \\
&\stackrel{B-G}{\leq} 401 \cdot \left( \frac{\int_0^1 \left( \frac{\sinh(s/r_k)}{r_k} \right)^{n-1} ds}{\int_0^{\frac{1}{2}} \left( \frac{\sinh(s/r_k)}{r_k} \right)^{n-1} ds} - 1 \right) \\
&\leq C \quad \text{since } T < \infty \text{ and therefore } r_k \text{ is bounded.} \quad (1.75)
\end{aligned}$$

Next,  $|Rm(g(t_k))| \leq r_k^{-2}$  implies the scalar curvature bound  $R \leq n(n-1)r_k^{-2}$ , which yields

$$\begin{aligned}
\bar{R}_k(s) &\leq n(n-1) \quad \text{for } 0 \leq s \leq 1. \\
&\implies \mathcal{W}(g(t_k), f_k, r_k^2) \leq \text{const.} + c_k \\
&\stackrel{c_k \rightarrow -\infty}{\implies} \mathcal{W}(g(t_k), f_k, r_k^{-2}) \stackrel{k \rightarrow \infty}{\rightarrow} -\infty \\
&\implies \mu(g(t_k), r_k^2) \rightarrow -\infty.
\end{aligned}$$

Recall that, for any  $t < t_0$ ,  $\mu(g(t), t_0 - t)$  is nonincreasing in  $t$  by **Proposition 1.14**. Hence,

$$\begin{aligned}
\mu(g(0), t_k + r_k^2) &\leq \mu(g(t_k), r_k^2) \stackrel{k \rightarrow \infty}{\rightarrow} -\infty \\
&\implies \mu(g(0), t_k + r_k^2) \stackrel{k \rightarrow \infty}{\rightarrow} -\infty
\end{aligned}$$

Since  $T$  finite,  $t_k + r_k^2$  is uniformly bounded,  $t_k$  uniformly positive, contradicting to the fact that  $\mu(g(0), \tau)$  is a continuous function in  $\tau$ .

□

We can also prove another version of local collapsing theorem under weaker conditions, whose proof is presented in [HN14]:

**Theorem 1.17** ( $\kappa$ -noncollapsing). *Fix  $t \in [-T, 0]$ ,  $x \in M$ ,  $r > 0$ . Assume that*

$$\inf_{\rho \in (0, r)} \mu(g(-T), t + T + \rho^2) \geq -C, \quad \sup_{B_r(x, t)} R(g(t)) \leq Cr^{-2}. \quad (1.76)$$

*Defining  $\kappa = \exp(-2^{n+4} - 2C)$ , then it follows that*

$$|B_r(x, t)| \geq \kappa r^n. \quad (1.77)$$

*Proof.* We work in each  $t$ -slice. For  $\rho \in (0, r)$ , define a smooth Lipschitz function

$$\psi = \begin{cases} 1 & \text{in } B(x, \rho/2) \\ \text{linear in between} & \\ 0 & \text{outside } B(x, \rho). \end{cases}$$

Using the monotonicity of  $\mu$ , for  $\varphi = \psi / \|\psi\|_2$ , we have

$$\int \varphi^2 \log \varphi^2 \frac{dV}{|B(x, \rho)|} \stackrel{Jensen}{\geq} \left( \int \varphi^2 \frac{dV}{|B(x, \rho)|} \right) \cdot \log \left( \int \varphi^2 \frac{dV}{|B(x, \rho)|} \right) \quad (1.78)$$

$$= \frac{1}{|B(x, \rho)|} \cdot \log \left( \frac{1}{|B(x, \rho)|} \right) \quad (1.79)$$

which yields, from (1.69):

$$\log \left( \frac{1}{|B(x, \rho)|} \right) \leq \frac{16\tau}{\rho^2} \left( \frac{B(x, \rho)}{B(x, \rho/2)} - 1 \right) + \frac{C\tau}{\rho^2} - \frac{n}{2} \log(4\pi\tau) - n - \mu(g(-T), \tau + t + T) \quad (1.80)$$

for all  $\tau > 0$ . Now let  $\tau = \rho^2$ . Using our definition of  $\kappa$  we deduce that if we have for some small  $\rho$

$$|B(x, \rho/2)| \geq \kappa \left( \frac{\rho}{2} \right)^n \quad (1.81)$$

then

$$|B(x, \rho)| \geq \kappa \rho^n. \quad (1.82)$$

And as  $\kappa < |B_{\mathbb{R}^n}(1)|$ , theorem follows from the induction on  $\rho$ .  $\square$

#### 1.1.4 Geometric implications of monotonicity

For a given functional, it is natural to consider its critical points. For the functionals introduced above, these critical points correspond to smooth metrics on compact manifolds. Here, we explore different types of spaces that arise as critical points of various functionals.

##### 1. Gaussian

Consider  $\mathbb{R}^n$  with the standard metric  $g_{can}$ , constant in time. Fix  $t_0 > 0$ . Put  $\tau = t_0 - t$  and define

$$f(t, x) = \frac{|x|^2}{4\tau} + \frac{n}{2} \ln(4\pi\tau), \quad (1.83)$$

so we would have

$$e^{-f} = (4\pi\tau)^{-n/2} e^{-\frac{|x|^2}{4\tau}}, \quad (1.84)$$

which is precisely the standard forward heat kernel for  $\tau \in [0, t_0)$  (backward for  $t$ ). Via direct computation, we notice that  $(g_{can}, f(t, x))$  solves

$$\begin{cases} g_t = -2Ric_g \equiv 0 \\ f_t = -\Delta f - R + |\nabla f|^2 \end{cases} \quad (1.85)$$

and  $\int_{\mathbb{R}^n} f(t, x) dx \equiv 1$ . Moreover, we can check that

$$\mathcal{F}(t) = \int_{\mathbb{R}^n} |\nabla f|^2 e^{-f} dx = \frac{n}{2\tau}, \quad (1.86)$$

which is a nondecreasing function in  $t \in [0, t_0)$ .

##### 2. Breathers

Roughly speaking, breathers are special critical points to the Ricci flow. More precisely,

**Definition 1.18** (Steady breather). A **steady breather** is a solution to Ricci flow on an interval  $[t_1, t_2]$  that satisfies the equation  $g(t_2) = \varphi^*g(t_1)$  for some  $\varphi \in \text{Diff}(M)$ . In particular, steady gradient Ricci solitons are steady breathers.

**Definition 1.19** (Expanding breather). An **expanding breather** is a solution to Ricci flow on an interval  $[t_1, t_2]$  that satisfies the equation  $g(t_2) = c\varphi^*g(t_1)$  for some  $\varphi \in \text{Diff}(M)$  and  $c > 1$ . In particular, expanding gradient Ricci solitons are expanding breathers.

**Definition 1.20** (Shrinking breather). An **expanding breather** is a solution to Ricci flow on an interval  $[t_1, t_2]$  that satisfies the equation  $g(t_2) = c\varphi^*g(t_1)$  for some  $\varphi \in \text{Diff}(M)$  and  $c < 1$ . In particular, shrinking gradient Ricci solitons are shrinking breathers.

Applying the monotonicity of functionals that we defined, we have the following results for compact breathers:

**Proposition 1.21.** A compact steady breather is a gradient steady Ricci soliton.

*Proof.* We have  $\lambda(g(t_2)) = \lambda(\varphi^*g(t_1)) = \lambda(g(t_1))$ , implying  $\mathcal{F}(g(t), f(t)) \equiv \text{const.}$ , yielding  $R + \Delta f = 0$ .  $\square$

**Proposition 1.22.** A compact expanding breather is an expanding gradient Ricci soliton.

*Proof.* Identical to the steady case: utilize the monotonicity of  $\bar{\lambda}$ -functional.  $\square$

**Proposition 1.23.** A compact shrinking breather is a shrinking gradient Ricci soliton.

*Proof.* Put  $t_0 = \frac{t_2 - ct_1}{1-c}$ . Then if  $\tau_1 = t_0 - t_1$  and  $\tau_2 = t_0 - t_2$ , then  $\tau_2 = c\tau_1$ . Hence

$$\begin{aligned} \mu(g(t_2), \tau_2) &= \mu\left(\frac{\tau_2}{\tau_1}\varphi^*g(t_1), \tau_2\right) \\ &= \mu(\varphi^*g(t_1), \tau_1) = \mu(g(t_1), \tau_1). \end{aligned}$$

$\square$

### 3. Gradient steady Ricci soliton

For compact gradient steady Ricci solitons, we have a better result:

**Proposition 1.24.** A compact steady gradient Ricci soliton is Ricci flat.

*Proof.* This is the equality case of **Proposition 1.8**. Recall that the function  $f(t)$  is defined as the backward solution of heat equation with final value  $f(t_2) = \bar{f}(t_2)$ . Then such function  $f(t)$  must be the minimizer of  $\mathcal{F}(g(t), \cdot)$  for all  $t$ , i.e.

$$(-4\Delta + R)e^{-f/2} = \lambda e^{-f/2} \tag{1.87}$$

where  $\lambda(t) \equiv \text{const.}$  Equivalently:  $2\Delta f - |\nabla f|^2 + R = \lambda$ .

As  $R + \Delta f = \text{tr}(\text{Ric} + \nabla^2 f) = 0$  from (1.36), we have

$$\Delta f - |\nabla f|^2 = \lambda. \quad (1.88)$$

Then  $\Delta e^{-f} = \lambda e^{-f}$ . Integrating both sides:

$$\begin{aligned} 0 &= \int_M \Delta e^{-f} dV = -\lambda \int_M e^{-f} dV \\ &\implies \lambda = 0 \\ &\implies f \text{ is constant and } g \text{ is Ricci flat.} \end{aligned}$$

□

## 1.2 Hein-Naber's $\epsilon$ -regularity theorem

In this subsection, we state and prove Hein-Naber's  $\epsilon$ -regularity theorem in their 2014 paper *New Logarithmic Sobolev Inequalities and an  $\epsilon$ -Regularity Theorem for the Ricci Flow*. Throughout this section, assume that the pair

$$(M^n, g(t)), \quad t \in [-T, 0] \quad (1.89)$$

is a smooth Ricci flow.

### 1.2.1 Pointed entropies

Pointed Nash entropy is one of the main machineries of Hein-Naber's  $\epsilon$ -regularity theorem. We introduce the pointed Nash entropy and its properties.

**Definition 1.25** (Conjugate heat operator). *Given a (super) Ricci flow  $(M, g(t))_{t \in I}$  on a compact manifold, define the **heat operator***

$$\square := \partial_t - \Delta_{g_t}. \quad (1.90)$$

*A solution to the equation  $\square u = 0$  is called a **solution to the heat equation (coupled with the Ricci flow)**. Correspondingly, the operator*

$$\square^* := -\partial_t - \Delta_t - \frac{1}{2} \text{tr}(\partial_t g_t) \quad (1.91)$$

*is called the **conjugate heat operator**. Under Ricci flow, we have*

$$\square^* = -\partial_t - \Delta_t + R(t). \quad (1.92)$$

*For any  $x, y \in M$  and  $s, t \in I$ ,  $s < t$  we denote by  $K(x, t; y, s)$  the heat kernel of  $\square$ . That is, for fixed  $(y, s)$ , the function  $K(\cdot, \cdot; y, s)$  is a **heat kernel based at  $(y, s)$** , i.e.*

$$\square K(\cdot, \cdot; y, s) = 0, \quad \lim_{t \searrow s} K(\cdot, t; y, s) = \delta_y. \quad (1.93)$$

*For fixed  $(x, t)$ , the function  $K(x, t; \cdot, \cdot)$  is called a **conjugate heat kernel based at  $(x, t)$** , i.e.*

$$\square^* K(x, t; \cdot, \cdot) = 0, \quad \lim_{s \nearrow t} K(x, t; \cdot, s) = \delta_x. \quad (1.94)$$

**Remark 1.26.** *Note that, for any solution  $v$  to the **conjugate heat equation***

$$\square^* v = 0 \quad (1.95)$$

*we have*

$$\frac{d}{dt} \int_M v \, d\text{vol}_{g_t} = 0. \quad (1.96)$$

*Therefore*

$$\int_M K(x, t; \cdot, s) \equiv 1. \quad (1.97)$$

**Definition 1.27** (Pointed  $\mathcal{W}$ -entropy). *The **pointed  $\mathcal{W}$ -entropy at scale  $|s|$  based at  $x_0$**  is*

defined by

$$\mathcal{W}_{x_0}(s) := \mathcal{W}(g(s), f_{x_0}(\cdot, s), |s|) \quad (1.98)$$

where  $s \in [-T, 0)$ ,  $g(s)$  is a solution to Ricci flow, and  $f_{x_0}(\cdot, s)$  is defined by

$$K(x_0, 0; y, s) = (4\pi |s|)^{-n/2} \exp(-f_{x_0}(y, s)). \quad (1.99)$$

**Remark 1.28.** A few remarks regarding conjugate heat kernel:

- For convenience, denote  $K_{x_0}(y, s) = K(x_0, 0; y, s)$  for  $(x_0, 0) \in M \times [-T, 0]$  and

$$d\nu_{x_0}(y, s) = K_{x_0}(y, s) d\text{vol}_{g(s)}(y) \quad (1.100)$$

for the associated probability measures on  $M$ .

- The corrected flow equation (1.65) tells us that  $(4\pi\tau)^{-n/2}e^{-f}$  evolves by the conjugate heat equation associated to RF. Therefore, by applying the gradient formula (1.63), we have

$$\mathcal{W}_{x_0}(s) \leq 0, \quad \frac{d}{ds}\mathcal{W}_{x_0}(s) \geq 0, \quad \lim_{s \rightarrow 0} \mathcal{W}_{x_0}(s) = 0. \quad (1.101)$$

Now we use pointed  $\mathcal{W}$ -entropy and conjugate heat kernel to define the pointed Nash entropy.

**Definition 1.29** (Pointed Nash entropy). The **pointed Nash entropy** at  $x_0 \in M$ ,  $s \in [-T, 0)$ , is given by

$$\mathcal{N}_{x_0}(s) := \frac{1}{|s|} \int_s^0 \mathcal{W}_{x_0} dr = \int f_{x_0}(s) d\nu_{x_0}(s) - \frac{n}{2}. \quad (1.102)$$

Below we give a list of properties which pointed  $\mathcal{W}$  and  $\mathcal{N}$ -entropies satisfy without proof.

**Proposition 1.30.** The following hold for all  $x \in M$  and  $s \in [-T, 0)$ :

1.  $\mu(g(-T), T) \leq \mathcal{W}_x(s) \leq 0$ ,
2.  $\mathcal{W}_x(s) = - \int_s^0 2|r| \int \left| Ric + \nabla^2 f_x - \frac{1}{2|r|}g \right|^2 d\nu_x(r) dr$ ,
3.  $\mathcal{W}_x(s) \leq \mathcal{N}_x(s) \leq 0$ ,
4.  $\frac{d}{ds}\mathcal{N}_x(s) = \frac{1}{|s|}(\mathcal{N}_x(s) - \mathcal{W}_x(s)) \geq 0$ ,
5.  $\mathcal{N}_x(s) = - \int \log K_x(s) d\nu_x(s) - \frac{n}{2}(1 + \log 4\pi |s|) = \int f_x(s) d\nu_x(s) - \frac{n}{2}$ ,
6.  $\mathcal{N}_x(s) = - \int_s^0 2|r| \left(1 - \frac{r}{s}\right) \int \left| Ric + \nabla^2 f_x - \frac{1}{2|r|}g \right|^2 d\nu_x(r) dr$ .

### 1.2.2 A log-Sobolev inequality and heat kernel estimate

To prove the Lipschitz continuity of the pointed Nash entropy, all we need to do is to estimate the integral

$$\mathcal{I} = \int |\nabla_x (f_x(y, s) K_x(y, s))| d\text{vol}_{g(s)}(y).$$

Expressing  $K_x(y, s)$  in terms of  $f_x$  and writing  $d\nu = d\nu_x(s)$ , we find that

$$\mathcal{I} = \int |\nabla_x f_x - f_x \nabla_x f_x| d\nu \leq \|\nabla_x f_x\|_{L^2(d\nu)} \left(1 + \|f_x\|_{L^2(d\nu)}\right).$$

Indeed, the following logarithmic-Sobolev (Poincaré) inequality will give a bound on the  $\|f_x\|_{L^2(d\nu)}$  term:

**Theorem 1.31** (Log-Sobolev inequality). *Let  $(M, g(t))$  be a Ricci flow as in (1.89). Fix a point  $x_0 \in M$  in the final time slice and let  $s \in [-T, 0)$ .*

1. (Log-Poincaré) *For all  $\varphi \in C_0^\infty(M)$  with  $\int \varphi d\nu_{x_0}(s) = 0$ ,*

$$\int \varphi^2 d\nu_{x_0}(s) \leq 2|s| \int |\nabla \varphi|_{g(s)}^2 d\nu_{x_0}(s). \quad (1.103)$$

*Equality holds if and only if either  $\varphi \equiv 0$  or  $(M, g(t)) = (M', g'(t)) \times (\mathbb{R}, dz^2)$  isometrically for all  $t \in [s, 0]$  with  $z(x_0) = 0$  and  $\varphi = \lambda z$  for some  $\lambda \in \mathbb{R} \setminus \{0\}$ .*

2. (Log-Sobolev) *For all  $\varphi \in C_0^\infty(M)$  with  $\int \varphi d\nu_{x_0}(s) = 1$ ,*

$$\int \varphi^2 \log \varphi^2 d\nu_{x_0}(s) \leq 4|s| \int |\nabla \varphi|_{g(s)}^2 d\nu_{x_0}(s). \quad (1.104)$$

*Equality holds if and only if either  $\varphi \equiv 1$  or  $(M, g(t)) = (M', g'(t)) \times (\mathbb{R}, dz^2)$  isometrically for all  $t \in [s, 0]$  with  $z(x_0) = 0$  and  $\varphi = \exp(\lambda z - 2\lambda^2 |s|)$  for some  $\lambda \in \mathbb{R} \setminus \{0\}$ .*

To prove the above inequalities, the following “heat kernel homotopy” principle is simple but essential:

**Lemma 1.32** (“Heat kernel homotopy”). *Given  $s \leq t$  in  $[-T, 0]$ , write  $P_{st}u$  for the evolution of  $u \in C_0^\infty(M)$  from time  $s$  to time  $t$  under the forward heat equation coupled to Ricci flow, i.e.*

$$(P_{st}u)(x) = \int u(y) K(x, t; y, s) d\text{vol}_{g(s)}(y). \quad (1.105)$$

$P_{st}$  satisfies the following “homotopy”:

1. *If  $U_t$  are smooth functions depending on  $t \in [-T, 0]$ , then*

$$\frac{d}{dt} P_{t0} U_t = P_{t0} \square_t U_t. \quad (1.106)$$

2. *Let  $u \in C_0^\infty(M)$  and  $u_t := P_{st}u$  so that  $\square_t u_t = 0$ . Fix smooth functions (at least  $C^2$ )  $\varphi, \psi : \mathbb{R} \rightarrow \mathbb{R}$ . Write  $U_t = \varphi(u_t)$ ,  $V_t = \psi(u_t) |\nabla u_t|_{g(t)}^2$ . Then we have*

$$\square U_t = -\varphi''(u_t) |\nabla u_t|_{g_t}^2 \quad (1.107)$$

$$\square V_t = -2\psi(u_t) |\nabla^2 u_t|_{g(t)}^2 - 4\psi'(u_t) \langle \nabla^2 u_t, du_t \otimes du_t \rangle - \psi''(u_t) |du_t \otimes du_t|_{g(t)}^2 \quad (1.108)$$

*Proof.* (1.106) follows from the representation formula (1.105); (1.107) and (1.108) follows from direct computation and the properties of conjugate heat kernel.  $\square$

*Proof of Theorem 1.31.* We can rewrite inequalities (1.103) and (1.104) in a more convenient way. Writing  $d\nu = d\nu_{x_0}(s)$ , it suffices to prove that for all  $u \in C_0^\infty(M)$ , with  $u \geq 0$  in the second case,

$$\int u^2 d\nu - \left( \int u d\nu \right)^2 \leq 2|s| \int |\nabla u|^2 d\nu, \quad (1.109)$$

$$\int u \log u d\nu - \left( \int u d\nu \right) \log \left( \int u d\nu \right) \leq 2|s| \int \frac{|\nabla u|^2}{u} d\nu. \quad (1.110)$$

Combining the homotopy formula (1.106) and (1.107), defining  $\varphi(x) = x^2$  or  $= x \log x$ , we have

$$\int \varphi(u) d\nu - \varphi \left( \int u d\nu \right) = - \int_s^0 \frac{d}{dt} P_{t0}(\varphi(P_{st}u))(x_0) dt \quad (1.111)$$

$$= \int_s^0 P_{t0}(\varphi''(P_{st}u) |\nabla P_{st}u|_{g(t)}^2) dt. \quad (1.112)$$

Then we estimate the above integrand by (1.108), defining  $\psi = \varphi''$ , replacing  $t, 0$  by  $r, t$  for a new variable  $r \in [s, t]$ . Using the homotopy formula (1.106) and (1.107)-(1.108), we have

$$|\nabla P_{st}u|_{g(t)}^2 = P_{st} \left( (|\nabla u|_{g(s)}^2) \right) - 2 \int_t^s P_{rt} |\nabla^2 P_{sr}u|_{g(r)}^2 dr, \quad (1.113)$$

$$\frac{|\nabla P_{st}u|_{g(t)}^2}{P_{st}u} = P_{st} \left( \frac{|\nabla u|_{g(s)}^2}{u} \right) - 2 \int_t^s P_{rt} \left( (P_{sr}u) |\nabla^2 \log P_{sr}u|_{g(r)}^2 \right). \quad (1.114)$$

Substituting (1.113) and (1.114) into (1.112), we obtain the desired log-Sobolev inequalities. Equality occurs if and only if  $|\nabla^2 P_{sr}u|$  and  $|\nabla^2 \log P_{sr}u|$  are 0, i.e. either  $u$  is constant or the flow lines of  $\nabla u$  and  $\nabla \log u$  split off as isometric  $\mathbb{R}$ -factors.  $\square$

As the above Poincaré inequality will give us the estimate of  $\|f_x\|_{L^2(d\nu)}$  term in terms of the norm of its derivative, we are left with an estimate on the derivative of  $f_x$ . To do that, [HN14] cites two estimates on our heat kernel, both proven by Qi S. Zhang:

**Theorem 1.33.** *Define two auxillary quantities*

$$\rho := \|R[g(-T)^-]\|_{L^\infty}, \quad \mu := \inf_{\tau \in (0, 2T)} \mu(g(-T), \tau). \quad (1.115)$$

Let  $u : M \times [t_1, t_2] \rightarrow \mathbb{R}^+$  with  $[t_1, t_2] \subset [-T, 0]$  be a smooth positive solution to

$$\frac{\partial u}{\partial t} = \Delta_{g(t)} u. \quad (1.116)$$

For each  $t \in [t_1, t_2]$  we then have

$$\|u(t)\|_{L^\infty} \leq (4\pi(t - t_1))^{-n/2} \exp(\rho(t - t_1) - \mu) \|u(t_1)\|_{L^1}. \quad (1.117)$$

**Theorem 1.34** (Logarithmic Harnack inequality). *For any positive solution  $u$  of the forward heat*

equation coupled to Ricci flow as in Theorem 1.33, we have a spatial Harnack estimate

$$\left| \nabla \sqrt{\log \frac{\sup u}{u}} \right| \leq \frac{1}{\sqrt{t-t_1}}. \quad (1.118)$$

From the above estimates, we can easily derive the much needed derivative bound on  $f_x$  as follows.

**Corollary 1.35** (Derivative estimate on  $f_x$ ). *For each  $C > 0$  there exists  $C' = C'(n, C) > 0$  such that if*

$$R[g(s)] \geq -\frac{C}{|s|}, \quad \inf_{\tau \in (0, 2|s|)} \mu(g(s), \tau) \geq -C \quad (1.119)$$

and if we write  $K(x, 0; y, s) = (4\pi |s|)^{-n/2} \exp(-f_x(y, s))$  as in (1.99), then

$$|\nabla_x f_x|^2 \leq \frac{C'}{|s|} (C' + f_x). \quad (1.120)$$

*Proof.* Fix  $y, s$  and let  $u(x, t) = K(x, t; y, s)$ , so that  $u$  solves the forward heat equation. Then, with  $T = |s|$  and  $[t_1, t_2] = [s + \epsilon, 0]$  for  $\epsilon \rightarrow 0$ , we can apply **Theorem 1.33** to obtain

$$\bar{u} := \sup_{[\frac{s}{2}, 0] \times M} u \leq C' |s|^{-n/2}. \quad (1.121)$$

Applying **Theorem 1.34** with  $[t_1, t_2] = [\frac{s}{2}, 0]$  yields

$$|\nabla_x f_x|^2 \leq \frac{C'}{|s|} \log \frac{\bar{u}}{u} \quad \text{at } (x, 0). \quad (1.122)$$

□

As we have commented previously, first we show the Lipschitz continuity of the pointed Nash entropy. By **Corollary 1.35**, we have

$$\mathcal{I} \leq \|\nabla_x f_x\|_{L^2(d\nu)} \left( 1 + \|f_x\|_{L^2(d\nu)} \right) \leq C' |s|^{-\frac{1}{2}} \left( 1 + \int |f_x|^2 d\nu \right). \quad (1.123)$$

Now we are ready to prove the following important result.

**Theorem 1.36** (Lipschitz continuity of  $\mathcal{N}_x$ ). *Let  $(M, g(t))_{-T \leq t \leq 0}$  be a solution to the Ricci flow. Suppose for each  $C > 0$  there exists a  $C' = C'(n, C) > 0$  such that the following holds:*

$$R[g(s)] \geq -\frac{C}{|s|}, \quad \inf_{\tau \in (0, 2|s|)} \mu(g(s), \tau) \geq -C \quad (1.124)$$

for some  $s \in [-T, 0)$ . Then the mapping

$$(M, g(0)) \ni x \mapsto f_x(s) K_x(s) \in L^1(M, d\text{vol}_{g(s)}) \quad (1.125)$$

is globally  $C' |s|^{-1/2}$ -Lipschitz. In particular, this implies that

$$|\mathcal{N}_{x_1}(s) - \mathcal{N}_{x_2}(s)| \leq C' |s|^{-1/2} d_{g(0)}(x_1, x_2). \quad (1.126)$$

*Proof.* From **Proposition 1.30**, we have  $\int f_x d\nu \in [\frac{n}{2} - C, \frac{n}{2}]$ . Notice that the inequality  $\mathcal{N}_x(s) \geq \mathcal{W}_x(s)$  is equivalent to

$$\int \left( |\nabla f_x|^2 + R \right) d\nu \leq \frac{n}{2|s|}. \quad (1.127)$$

Using the log-Poincaré inequality in **Theorem 1.31**, we conclude that

$$\int |f_x|^2 d\nu \leq 2|s| \int |\nabla f_x|^2 d\nu + \left( \int f_x d\nu \right)^2 \leq C' |s|^{-\frac{1}{2}} \left( 1 + 2n + \frac{n^2}{4} \right), \quad (1.128)$$

which is the desired estimate.  $\square$

### 1.2.3 Proof of the $\epsilon$ -regularity theorem

First we give the statement of our  $\epsilon$ -regularity theorem.

**Definition 1.37** (Regularity scale). *For a Ricci flow  $(M^n, g(t))$  as in (1.89):*

1. *Given  $(x, t) \in M \times [-T, 0]$  and  $r > 0$  with  $-T \leq t - r^2$ , we define the **parabolic ball***

$$P_r(x, t) := B_r(x, t) \times (t - r^2, t]. \quad (1.129)$$

2. *Given  $(x, t) \in M \times [-T, 0]$ , we define the **regularity scale***

$$r_{|Rm|}(x, t) := \sup \{ r > 0 : \sup_{P_r(x, t)} |Rm| \leq r^{-2} \}. \quad (1.130)$$

**Theorem 1.38** ( $\epsilon$ -regularity theorem). *For each  $C > 0$  there exists an  $\epsilon = \epsilon(n, C) > 0$  such that the following holds: let  $(M^n, g(t))$  be any Ricci flow as in (1.89) such that*

$$R[g(s)] \geq -\frac{C}{|s|}, \quad \inf_{\tau \in (0, 2|s|)} \mu(g(s), \tau) \geq -C \quad (1.131)$$

*for some  $s \in [-T, 0)$ . If the pointed entropy satisfies*

$$\mathcal{W}_{x_0}(s) \geq -\epsilon \quad (1.132)$$

*for some point  $x_0$  in the 0-time slice, then we have*

$$r_{|Rm|}(x_0, 0)^2 \geq \epsilon |s|. \quad (1.133)$$

We prove a proposition which our main theorem follows directly from:

**Proposition 1.39.** *Suppose  $(M^n, g(t))$  satisfies (1.131). For a point  $y \in M$  on the 0-time slice, define*

$$t(y) := -\min\{T, r_{|Rm|}(y, 0)^2\}. \quad (1.134)$$

*There exists an  $\epsilon = \epsilon(n, C) > 0$  such that if we have*

$$\forall y \in B_\delta(x, 0), \mathcal{N}_{t(y)}(y) \geq -\epsilon \quad (1.135)$$

for some  $x \in M$  and  $0 < \delta \leq \sqrt{T}$ , then we also have

$$\forall y \in B_\delta(x, 0) : r_{|Rm|}(y, 0) \geq \epsilon \cdot d_{g(0)}(y, \partial B_\delta(x, 0)). \quad (1.136)$$

*Proof of Proposition 1.39.* We argue by contradiction.

- By parabolic rescaling, assume  $\delta = 1 \leq T$ . Also assume for some  $n, C > 0$  the proposition fails. Then we have for all  $i \in \mathbb{N}$  there exists a complete Ricci flow

$$(M_i^n, g_i(t), (x_i, 0)) \quad (1.137)$$

with  $t \in [-1, 0]$  and  $x_i \in M_i$  such that for each  $y \in B_1(x_i, 0)$  we have

$$\mathcal{N}_{t(y)}(y) \geq -\frac{1}{i}, \quad (1.138)$$

whereas any point  $y_i \in B_1(x_i, 0)$  minimizing the quantity

$$w(y) := \frac{r_{|Rm|}(y, 0)}{d_{g_i(0)}(y, \partial B_1(x_i, 0))} \quad (1.139)$$

must necessarily satisfy

$$0 < w(y_i) \leq \frac{1}{i}. \quad (1.140)$$

- Choose any such  $y_i$ , define  $r_i := r_{|Rm|}(y_i, 0)$ , and rescale the flow:

$$(\widetilde{M}_i, \widetilde{g}_i(t), (\widetilde{x}_i, 0)), \quad \widetilde{g}_i(t) := r_i^{-2} g_i(r_i^2 t), \quad t \in [-r_i^{-2}, 0]. \quad (1.141)$$

Then  $r_{|Rm|}(\widetilde{x}_i, 0) = 1$  and  $\widetilde{y}_i$  moves away from the boundary in that

$$d_i := \frac{1}{2} d_{\widetilde{g}_i(0)}(\widetilde{y}_i, \partial B_{\frac{1}{r_i}}(\widetilde{x}_i, 0)) \geq \frac{i}{2}. \quad (1.142)$$

On the other hand, since  $y_i$  minimizes  $w$ , we find that

$$\widetilde{y} \in B_{d_i}(\widetilde{y}_i, 0) \implies r_{|Rm|}(\widetilde{y}, 0) \geq \frac{1}{2}. \quad (1.143)$$

The assumption (1.131) and **Theorem 1.17** ( $\kappa$ -noncollapsing), we have

$$\text{Vol}_{\widetilde{g}_i(0)}(B_1(\widetilde{y}, 0)) \geq \kappa(n, C). \quad (1.144)$$

Therefore, we have uniform smooth bounds on the geometry of the parabolic ball  $P_{\frac{1}{4}}(\widetilde{y}, 0)$  for all  $\widetilde{y} \in B_{d_i}(\widetilde{y}_i, 0)$ , and uniform noncollapsing at scale 1 on  $B_{d_i}(\widetilde{y}_i, 0)$ . This allows us to pass to a subsequence and derive a pointed  $C^\infty$  limit

$$(\widetilde{M}_i, \widetilde{g}_i(t), (\widetilde{y}_i, 0)) \rightarrow (\widetilde{M}_\infty, \widetilde{g}_\infty(t), (\widetilde{y}_\infty, 0)). \quad (1.145)$$

The limiting flow exists for all times  $-\frac{1}{16} \leq t \leq 0$  and is complete with bounded curvature

satisfying

$$r_{|Rm|}(\widetilde{y}_\infty, 0) = 1. \quad (1.146)$$

- To derive contradiction, we will need our heat kernel estimates. Write

$$K(\widetilde{y}_i, 0; \widetilde{y}, t) := (4\pi |t|)^{-\frac{n}{2}} e^{-f_i(\widetilde{y}, t)}. \quad (1.147)$$

Using the lower bounds on Perelman's reduced length function  $l$  plus **Theorem 1.33**, we have the following inequality: for all  $t \in [-1, 0)$ ,  $\widetilde{y} \in \widetilde{M}_i$  in the rescaled flows  $(\widetilde{M}_i, \widetilde{g}_i(t), (\widetilde{y}_i, 0))$ ,

$$(4\pi |t|)^{-\frac{n}{2}} \exp(-l_{\widetilde{y}_i}^{[t]}(\widetilde{y})) \leq K(\widetilde{y}_i, 0; \widetilde{y}, t) \leq C'(n, C) |t|^{-\frac{n}{2}}. \quad (1.148)$$

From the above inequality and the regularity bound (1.143), for all  $t \in [-\frac{1}{16}, 0)$ ,

$$f_i(\widetilde{y}, t) \rightarrow f_\infty(\widetilde{y}, t) \quad (1.149)$$

smoothly on compact sets since we already have uniform derivative bounds for the heat kernels  $K_{\widetilde{y}_i}$ . Now using the assumption on pointed Nash entropy (1.138) and **Proposition 1.30**, we have

$$\int_{-\frac{1}{16}}^0 2|t|(1-16|t|) \int_{\widetilde{M}_i} \left| Ric(\widetilde{g}_i) + \nabla^2 f_i - \frac{\widetilde{g}_i}{2|t|} \right|^2 d\nu_{\widetilde{y}_i}(t) dt \leq \frac{1}{i}. \quad (1.150)$$

Then, letting  $i \rightarrow \infty$  and applying Fatou's lemma, we find that for all  $t \in [-\frac{1}{16}, 0)$ ,  $f_\infty$  is a soliton potential for the limit flow. Therefore the  $t$ -time slice of the limit flow is isometric to that of  $-\frac{1}{16}$ -time slice, scaled by  $16|t|$ . But then the only way for the curvature to remain uniformly bounded as  $t \rightarrow 0$  is if  $(\widetilde{M}_\infty, \widetilde{g}_\infty(t), (\widetilde{y}_\infty, 0))$  is flat for all  $t$ , which contradicts (1.146).  $\square$

*Proof of Theorem 1.38.* Define  $\delta = \min\{1, \epsilon_{1.39}/2C'_{1.36}\}$  and  $\epsilon = \frac{\delta}{2}\epsilon_{1.39}$ . WLOG assume  $-s = T = 1 \geq \delta$ . Then by **Theorem 1.36**,

$$\forall x \in B_\delta(x_0, 0) : \mathcal{N}_x(1) \geq -\epsilon_{1.39}. \quad (1.151)$$

From **Proposition 1.39**, it follows that  $r_{|Rm|}(x_0, 0) \geq \epsilon_{1.39}\delta \geq \epsilon$ , which completes the proof.  $\square$

For the sake of completeness, below we state Perelman's pseudolocality theorem without proof for later reference.

**Theorem 1.40** (Perelman's pseudolocality theorem). *For every  $\alpha > 0$  there exist  $\delta, \epsilon > 0$  with the following property. Suppose that*

- *we have a smooth pointed Ricci flow solution  $(M, (x_0, 0), g(\cdot))$  defined for  $t \in [0, (\epsilon r_0)^2]$ , such that each time slice is complete;*
- *for any  $x \in B_0(x_0, r_0)$  and  $\Omega \subset B_0(x_0, r_0)$ , we have  $R(x, 0) \geq -r_0^2$  and  $\text{vol}(\partial\Omega)^n \geq (1 - \delta)c_n \text{vol}(\Omega)^{n-1}$ , where  $c_n$  denotes the Euclidean isoperimetric constant.*

*Then  $|Rm|(x, t) < \alpha t^{-1} + (\epsilon r_0)^{-2}$  whenever  $0 < t \leq (\epsilon r_0)^2$  and  $d(x, t) = \text{dist}_t(x, x_0) \leq \epsilon r_0$ .*

## 2 Bamler 20a

Contributions of Bamler's work [Bam20a] to the regularity theory of Ricci flow are several; for me the most (geometrically) enlightening element of this paper is the concept of  $P^*$ -parabolic neighborhood – a geometric object which can better describe the regularity of Ricci flow in a more general setting than the conventional parabolic neighborhoods. Below notes on [Bam20a] intend to lead the readers – somewhat in a rush – to the introduction of the  $P^*$ -parabolic neighborhood, its geometric/analytic features, and its role in the proof of Bamler's new  $\epsilon$ -regularity theorem. Therefore proofs are often omitted (which you can look up directly in Bamler's paper). We wish to make visible connections between Bamler's argument and Hein-Naber's approach for their respective  $\epsilon$ -regularity theorems in the notes.

It is important to keep in mind that, unlike the usual parabolic neighborhood, Bamler's  $P^*$ -parabolic neighborhood is defined using  $W_1$ -Wasserstein distance. We begin this section of the notes with the introduction of  $W_1$ -distance and its behaviors under (super) Ricci flow.

Unless stated otherwise,  $M = M^n$  will denote a compact smooth manifold of dimension  $n$ ,  $\square = \partial_t - \Delta_{g_t}$  will denote the heat operator on a Riemannian manifold  $(M, g_t)$ , and  $I$  an interval on  $\mathbb{R}$ .

### 2.1 Monotonicity of $W_1$ -distance and variance under (S)RF

Wasserstein distance is a specific kind of distance assigned between two probability measures.

**Definition 2.1** ( $W_1$ -Wasserstein distance). *For two probability measures  $\mu_1, \mu_2$  defined on a complete Riemannian manifold  $(M, g)$ , the  $W_1$ -distance between  $\mu_1, \mu_2$  is given by*

$$d_{W_1}^g(\mu_1, \mu_2) := \sup_{\text{bounded } 1\text{-Lipschitz } f: M \rightarrow \mathbb{R}} \left( \int_M f d\mu_1 - \int_M f d\mu_2 \right). \quad (2.1)$$

Below monotonicity lemma conveys the advantages of using the  $W_1$ -distance over our usual Riemannian distance when studying (super) Ricci flow.

**Lemma 2.2.** *Let  $(M, g_t)_{t \in I}$  be a super Ricci flow on a compact manifold and denote by  $v_1, v_2 \in C^\infty(M \times I')$ ,  $I' \subset I$ , two non-negative solutions to the conjugate heat equation  $\square^* v_1 = \square^* v_2 = 0$  such that  $\int_M v_i(\cdot, t) dg_t = 1$  for all  $t \in I'$ ,  $i = 1, 2$ . Denote by  $\mu_{1,t}, \mu_{2,t}$  the associated probability measures with  $d\mu_{i,t} = v_i(\cdot, t) dg_t$ . Then*

$$I' \rightarrow [0, \infty], \quad t \mapsto d_{W_1}^{g_t}(\mu_{1,t}, \mu_{2,t}) \quad (2.2)$$

*is nondecreasing. Moreover, for any two points  $x_1, x_2 \in M$  and  $t_0 \in I$  we have for all  $t \leq t_0, t \in I$ ,*

$$d_{W_1}^{g_t}(\nu_{x_1, t_0; t}, \nu_{x_2, t_0; t}) \leq d_{t_0}(x_1, x_2). \quad (2.3)$$

*Proof.* Proof of this lemma relies on the Bochner formula. For a  $C^3$ -solution  $u$  to the heat equation  $\square u = 0$ , we have

$$\square |\nabla u|_t^2 = \partial_t |\nabla u|^2 - \Delta_t |\nabla u|^2 \stackrel{SRF}{\leq} g_t^{ik} g_t^{jl} \cdot 2R_{ij}(t) u_k u_l + 2g_t(\nabla \Delta_t u, \nabla u)$$

which yields

$$\square |\nabla u|^2 \leq 2\text{Ric}(\nabla u, \nabla u) + 2g(\nabla \Delta u, \nabla u) - \Delta |\nabla u|^2.$$

Then by Bochner formula, we have

$$\Delta |\nabla u|^2 \leq -2 |\text{Hess } u|^2 = -2 |\nabla^2 u|^2. \quad (2.4)$$

Since

$$\begin{aligned} \square |\nabla u|^2 &= 2 |\nabla u| \partial_t |\nabla u| - 2 |\nabla u| \Delta |\nabla u| - 2 |\nabla |\nabla u||^2 \\ &= 2 |\nabla u| (\square |\nabla u|) - 2 |\nabla |\nabla u||^2 \\ &\leq -2 |\nabla^2 u|^2 \end{aligned}$$

By Kato's inequality

$$|\nabla |\nabla u|| \leq |\nabla^2 u|, \quad (2.5)$$

we have

$$\square |\nabla u| \leq 0. \quad (2.6)$$

Applying the maximum principle, we deduce that if  $\underline{u(\cdot, t_1)}$  is  $L$ -Lipschitz for some  $t_1 \in I'$  and  $L \geq 0$ , then so is  $\underline{u(\cdot, t)}$  for  $t_1 \leq t \in I'$ .

Now let  $t_1 \leq t_2$ ,  $t_1, t_2 \in I'$  and consider a function  $f \in C^\infty(M)$  with  $|\nabla f|_{g_{t_1}} \leq 1$ . Let  $u \in C^\infty(M \times [t_1, t_2])$  be the solution to the heat equation with initial condition  $u(\cdot, t_1) = f$ . From the above argument, we have  $|\nabla u| \leq 1$  at all times. So

$$\begin{aligned} \int_M f d\mu_{1,t_1} - \int_M f d\mu_{2,t_1} &= \int_M u(\cdot, t_1) v_1(\cdot, t_1) dg_{t_1} - \int_M u(\cdot, t_1) v_2(\cdot, t_1) dg_{t_1} \\ &= \int_M u(\cdot, t_2) v_1(\cdot, t_2) dg_{t_2} - \int_M u(\cdot, t_2) v_2(\cdot, t_2) dg_{t_2} \\ &= \int_M u(\cdot, t_2) d\mu_{1,t_2} - \int_M u(\cdot, t_2) d\mu_{2,t_2} \leq d_{W_1}^{g_{t_2}}(\mu_{1,t_2}, \mu_{2,t_2}). \end{aligned}$$

Taking the supremum over all such  $f$  implies the first statement of **Lemma 2.2**. The second follows from the first by taking the limit  $t \nearrow t_0$ .  $\square$

Now we discuss the notion of variance and its relation to  $W_1$ -distance.

**Definition 2.3** (Variance). *The **variance** between two probability measures  $\mu_1, \mu_2$  on a Riemannian manifold  $(M, g)$  is defined as*

$$\text{Var}(\mu_1, \mu_2) := \int_M \int_M d^2(x_1, x_2) d\mu_1(x_1) d\mu_2(x_2).$$

*In the case  $\mu_1 = \mu_2 = \mu$ , we also write  $\text{Var}(\mu, \mu) = \text{Var}(\mu)$ ; when  $g = g_t$ , we may write  $\text{Var}_t$  to denote its dependence on  $t$ . Also, it's useful to note that  $\text{Var}(\delta_x, \delta_y) = d^2(x, y)$ .*

**Lemma 2.4.** *For any three probability measures  $\mu_1, \mu_2, \mu_3$  on a Riemannian manifold  $(M, g)$ , we*

have

$$\sqrt{\text{Var}(\mu_1, \mu_3)} \leq \sqrt{\text{Var}(\mu_1, \mu_2)} + \sqrt{\text{Var}(\mu_2, \mu_3)} \quad (2.7)$$

$$d_{W_1}^g(\mu_1, \mu_2) \leq \sqrt{\text{Var}(\mu_1, \mu_2)} \leq d_{W_1}^g(\mu_1, \mu_2) + \sqrt{\text{Var}(\mu_1)} + \sqrt{\text{Var}(\mu_2)}. \quad (2.8)$$

Using the notion of variance, we define something called an “ $H_n$ -center” of a point.

**Definition 2.5** ( $H_n$ -center). *Define the constant*

$$H_n := \frac{(n-1)\pi^2}{2} + 4.$$

A point  $(z, t) \in M \times I$  is called an  $H_n$ -center of a point  $(x_0, t_0) \in M \times I$  if  $t \leq t_0$  and

$$\text{Var}_t(\delta_z, \nu_{x_0, t_0; t}) \leq H_n(t_0 - t). \quad (2.9)$$

**Remark 2.6.** By Lemma 2.4, we know that the bound (2.9) implies

$$d_{W_1}^{g_t}(\delta_z, \nu_{x_0, t_0; t}) \leq \sqrt{\text{Var}_t(\delta_z, \nu_{x_0, t_0; t})} \leq \sqrt{H_n(t_0 - t)}.$$

**Remark 2.7.** As a matter of fact, the constant  $H_n$  originates from the following inequality in the barrier sense:

$$(\partial_t - \Delta_x - \Delta_y)d_t^2(x, y) \geq -H_n. \quad (2.10)$$

Just as the  $W_1$ -distance, we know that the variance between two conjugate heat kernel measures also satisfies certain monotonicity property.

**Lemma 2.8.** *Consider two non-negative solutions  $v_1, v_2 \in C^\infty(M)$  to the conjugate heat equation  $\square^* v_i = 0$  with  $\int_M v_i dg_t = 1$ ,  $i = 1, 2$ , and let  $\mu_{i,t} = v_i dg_t$  be the corresponding probability measures. Then the map*

$$t \mapsto \text{Var}_t(\mu_{1,t}, \mu_{2,t}) + H_n t$$

*is non-decreasing. In particular, taking the conjugate heat kernel measures  $\nu_{x_i, t_i; t}$  for two points  $(x_i, t_i) \in M \times I$ ,  $i = 1, 2$ , then we have*

$$t \mapsto \text{Var}_t(\nu_{x_1, t_1; t}, \nu_{x_2, t_2; t}) + H_n t$$

*is non-decreasing for  $t \leq \min\{t_1, t_2\}$ . For  $t_0 = t_1$ , we obtain*

$$\text{Var}_t(\nu_{x_1, t_0; t}, \nu_{x_2, t_0; t}) \leq d_{t_0}^2(x_1, x_2) + H_n(t_0 - t), \quad \text{Var}_t(\nu_{x_1, t_0}(t)) \leq H_n(t_0 - t). \quad (2.11)$$

All of the above monotonicity lemmas amount to the following essential integral Gaussian bound on the metric  $t$ -ball centered at an  $H_n$ -center.

**Theorem 2.9.** *If  $(z, t)$  is an  $H_n$ -center of  $(x_0, t_0)$ , then for all  $r \geq 0$  and  $\epsilon > 0$ ,*

$$\begin{aligned}\nu_{x_0, t_0; t}(M \setminus B(z, t, r)) &= \int_{M \setminus B(z, t, r)} K(x_0, t_0; \cdot, t) dg_t \\ &\leq 2 \exp \left( -\frac{(r - \sqrt{2H_n(t_0 - t)})_+^2}{8(t_0 - t)} \right) \\ &\stackrel{\exists C(\epsilon)}{\leq} C(\epsilon) \exp \left( -\frac{r^2}{(8 + \epsilon)(t_0 - t)} \right).\end{aligned}$$

*Proof.* First, we prove a Gaussian integral bound of more general type, introduced in [HN14].

Fix a smooth function  $F$  on  $M$  with

$$\int_M F d\nu = 0, \quad |\nabla F| \leq 1$$

where  $\nu = \nu_{x_0, t_0; t}$ . Define a Laplace-type transform by

$$U(\lambda) := \frac{1}{\lambda} \log \int_M e^{\lambda F} d\nu = O(\lambda) \text{ as } \lambda \rightarrow 0. \quad (2.12)$$

Now the log-Sobolev inequality (1.104) applied to  $\phi^2 := e^{\lambda F} / \int_M e^{\lambda F} d\nu$  gives us

$$\frac{dU}{d\lambda} \leq |t|$$

for all  $\lambda > 0$ , together with the bound

$$\int_M e^{\lambda F} d\nu \leq e^{|t|\lambda^2}.$$

Apply the above inequality to two test functions

$$F := \pm \left( G - \int_M G \right), \quad G(y) = \text{dist}(y, B)$$

for two subsets  $A, B \subset M$ . Then we immediately obtain

$$e^{\lambda \text{dist}(A, B)} \nu(A) \nu(B) \leq \int_A \int_B e^{\lambda(F(y_1) - F(y_2))} d\nu(y_1) d\nu(y_2) \leq e^{2|t|\lambda}. \quad (2.13)$$

Optimizing over  $\lambda$ , we have

$$\nu(A) \nu(B) \leq \exp \left[ -\frac{\text{dist}_t(A, B)^2}{8|t|} \right]. \quad (2.14)$$

The above type of argument is the so-called ‘‘Herbst argument’’ in metric measure theory.

Now let  $B_0 := B(x, t, \sqrt{AH_n(t_0 - t)})$ . Using the inequality (2.11), we obtain that

$$\nu_{x_0, t_0; t}(M \setminus B_0) \leq \frac{1}{AH_n(t_0 - t)} \int_{M \setminus B_0} d_t^2(z, \cdot) d\nu_{x_0, t_0; t} \leq \frac{1}{AH_n(t_0 - t)} \text{Var}_t(\nu_{x_0, t_0; t}) \leq \frac{1}{A}. \quad (2.15)$$

Taking  $A = M \setminus B(z, t, r)$  and  $B = B(z, t, \sqrt{2H_n(t_0 - t)})$  in the inequality (2.14) and directly

applying the bound (2.15), we obtain the first bound in **Theorem 2.9**. The second bound in **Theorem 2.9** is a direct consequence of the first.  $\square$

Later in our notes, we will explore the geometry of time-slices of  $P^*$ -neighborhoods. And by then, we will employ **Theorem 2.9** as a tool for guaranteeing the non-collapsedness of “”. Also it is important to note that, albeit our notes will not give a very detailed account on it, Gaussian integral bound such as **Theorem 2.9** plays an important role when proving the upper bounds on the heat kernel and its gradient. (TBC)

## 2.2 Gradient estimates on heat kernel and bounds on Nash entropy

In this subsection, we list (and only list: the proofs often involve very messy computations; gradient estimates on heat kernels alone might deserve a separate stack of notes) certain results regarding the gradient estimates on the heat kernel. As we have witnessed in Hein-Naber’s paper, e.g. **Theorem 1.33** and **Theorem 1.34**, gradient estimates on heat kernel are particularly handy for obtaining the bounds on Nash entropy. Of course, then using the stated heat kernel estimates, we can prove several important improved bounds on the pointed Nash entropy.

Unless stated otherwise, we assume  $(M, g_t)_{t \in I}$  is a super Ricci flow on a compact manifold  $M$ .

First, there is a sharp gradient bound on  $\Phi_t(x) := \Phi(t^{-1/2}x) : \mathbb{R} \rightarrow (0, 1)$ , a solution to the 1-dimensional heat equation with initial value  $\chi_{[0, \infty)}$ , which satisfies

$$\Phi'(x) = (4\pi)^{-1/2} e^{-x^2/4}, \quad \lim_{x \rightarrow -\infty} \Phi(x) = 0; \quad \lim_{x \rightarrow \infty} \Phi(x) = 1.$$

**Theorem 2.10.** *Consider a solution  $u \in C^\infty(M \times [t_0, t_1])$ ,  $[t_0, t_1] \in I$ , to the heat equation  $\partial_t u = \Delta_t u$  on the super Ricci flow  $(M, g_t)_{t \in I}$  and let  $T \geq 0$ . Suppose that  $u$  only takes values in  $(0, 1)$  and suppose that*

$$|\nabla(\Phi_T^{-1}(u(\cdot, t_0)))| \leq 1 \text{ if } T > 0.$$

*Then*

$$|\nabla(\Phi_{T+t-t_0}^{-1}(u(\cdot, t_0)))| \leq 1 \text{ for all } t \in [t_0, t_1].$$

*Proof.* Follows from heat equation, Bochner formula, and maximum principle. Sharpness: consider the trivial Ricci flow on  $\mathbb{R}$  and  $u_t = \Phi_t$ . Taking the Cartesian product with any other super Ricci flow yields sharpness in all dimensions.  $\square$

Subtle use of the above bound can deduce the following important integral bounds on the gradient of the heat kernel, which is reminiscing of Zhang’s logarithmic Harnack inequality (1.118):

**Theorem 2.11** (Integral bounds on the gradient of  $K$ ). *Let  $[s, t] \subset I$  and write  $d\nu := d\nu_{x,t;s} = K(x, t; \cdot, s) dg_s$ . Then for any  $1 \leq p < \infty$*

$$(t-s)^{p/2} \int_M \left( \frac{|\nabla_x K(x, t; ds)|}{K(x, t; \cdot, s)} \right)^p d\nu \leq C(n, p). \quad (2.16)$$

Moreover, for any measurable subset  $X \subset M$  we have:

$$(t-s)^{p/2} \int_X \left( \frac{|\nabla_x K(x, t; ds)|}{K(x, t; \cdot, s)} \right)^p d\nu \leq C(n, p) \nu(X) (-\log(\nu(X))/2)^{p/2}. \quad (2.17)$$

In particular, taking  $p = 2$ , then we can take  $C(n, p) = \frac{n}{2}$  and we even have for any  $v \in T_x M$  with  $|v|_t = 1$ :

$$(t-s) \int_M \left( \frac{\partial_v K(x, t; \cdot, s)}{K(x, t; \cdot, s)} \right)^2 d\nu \leq \frac{1}{2}. \quad (2.18)$$

With conditions on the lower bound of the scalar curvature  $R$ , which is often available on a Ricci flow setting, we have the following  $L^\infty$  bound for  $K(x, t; y, s)$  and  $\frac{|\nabla_x K|}{K}$  that only depend on  $t, s$ , and the pointed Nash entropy  $\mathcal{N}_{x,t}$ .

**Theorem 2.12** ( $L^\infty$ -bound on  $K$ ). *If  $[s, t] \subset I$  and  $R \geq R_{\min}$  on  $M \times [s, t]$ , then*

$$K(x, t; y, s) \leq \frac{C(R_{\min}(t-s))}{(t-s)^{n/2}} \exp(-\mathcal{N}_{x,t}(t-s)), \quad (2.19)$$

where we may choose  $C(R_{\min}(t-s)) = C_0 \cdot (n - R_{\min}(t-s))^{n/2}$  for some dimensional constant  $C_0 = C_0(n) < \infty$ .

**Theorem 2.13** (Gradient bound on  $K$ ). *If  $[s, t] \subset I$  and  $R \geq R_{\min}$  on  $M \times [s, t]$ , then there are constants  $C(R_{\min}(t-s)), C_0(R_{\min}(t-s)) < \infty$  such that*

$$\frac{|\nabla_x K|(x, t; y, s)}{K(x, t; y, s)} \leq \frac{C}{(t-s)^{1/2}} \sqrt{\log \left( \frac{C_0 \exp(-\mathcal{N}_{x,t}(t-s))}{(t-s)^{n/2} K(x, t; y, s)} \right)}. \quad (2.20)$$

Indeed, the above two pointwise bounds rely on the properties and estimates which the pointed Nash entropy  $\mathcal{N}_{x,t}$  satisfy. Below we derive several bounds of  $\mathcal{N}_{x,t}$  using the integral bound of the heat kernel provided in **Theorem 2.11**.

Recall Hein-Naber's definition of the pointed Nash entropy  $\mathcal{N}_{x_0}(s)$ ,  $s \in (-T, 0]$ . We shall make a slight modification to the definition/notation of the pointed Nash entropy as follows:

**Definition 2.14** (Pointed Nash entropy). *Let  $(M, g_t)_{t \in I}$  be a Ricci flow on a compact manifold and consider a conjugate heat kernel measure  $d\nu_{x_0, t_0} = (4\pi\tau)^{-n/2} e^{-f} dg_{t_0} = K(x_0, t_0; \cdot, \cdot) dg_{t_0}$  based at some point  $(x_0, t_0) \in M \times I$ , where  $\tau = t_0 - t$ . The **pointed Nash entropy at  $(x_0, t_0)$**  is defined as*

$$\mathcal{N}_{x_0, t_0}(\tau) := \mathcal{N}[g_{t_0-\tau}, f_{t_0-\tau}, \tau] = \int_M f_{t_0-\tau} d\nu_{x_0, t_0-\tau} - \frac{n}{2}$$

where

$$(4\pi\tau)^{-n/2} e^{-f_{t_0-\tau}} = K(x_0, t_0 - \tau; \cdot, \cdot).$$

We set  $\mathcal{N}_{x_0, t_0}(0) := 0$ . For  $s < t_0$ ,  $s \in I$ , we also write

$$\mathcal{N}_s^*(x_0, t_0) := \mathcal{N}_{x_0, t_0}(t_0 - s).$$

With a lower bound on  $R$ , the pointed Nash entropy  $\mathcal{N}_{x_0, t_0}$  enjoys richer properties.

**Proposition 2.15** (Properties of pointed Nash entropy). *The expression  $\mathcal{N}_{x_0, t_0}(\tau)$  is continuous for  $\tau \geq 0$  and if  $R(\cdot, t_0 - \tau) \geq R_{\min}$ , then for  $\tau > 0$*

$$\frac{d}{d\tau}(\tau \mathcal{N}_{x_0, t_0}(\tau)) = \mathcal{W}[g_{t_0 - \tau}, f_{t_0 - \tau}, \tau] \leq 0, \quad (2.21)$$

$$\frac{d^2}{d\tau^2}(\tau \mathcal{N}_{x_0, t_0}(\tau)) = -2\tau \int_M \left| Ric_{t_0 - \tau} + \nabla^2 f_{t_0 - \tau} - \frac{1}{2\tau} g_{t_0 - \tau} \right|^2 d\nu_{t_0 - \tau} \leq 0, \quad (2.22)$$

$$-\frac{n}{2\tau} + R_{\min} \leq \frac{d}{d\tau} \mathcal{N}_{x_0, t_0}(\tau) \leq 0. \quad (2.23)$$

If  $\tau_1 \leq \tau_2$  and  $R \geq R_{\min}$  on  $M \times [t_0 - \tau_2, t_0 - \tau_1]$ , then

$$\mathcal{N}_{x_0, t_0}(\tau_1) - \frac{n}{2} \log \left( \frac{\tau_2}{\tau_1} \left( 1 - \frac{2}{n} R_{\min}(\tau_2 - \tau_1) \right) \right) \leq \mathcal{N}_{x_0, t_0}(\tau_2) \leq \mathcal{N}_{x_0, t_0}(\tau_1). \quad (2.24)$$

*Proof.* Direct computation.  $\square$

What follows next are the derivative bounds of  $\mathcal{N}$  under a lower scalar curvature bound.

**Theorem 2.16.** *If  $R(\cdot, s) \geq R_{\min}$  for some fixed  $s \in I$ , then on  $M \times (I \cap (s, \infty))$  we have*

$$|\nabla \mathcal{N}_s^*| \leq \left( \frac{n}{2(t-s)} - R_{\min} \right)^{1/2}, \quad -\frac{n}{2(t-s)} \leq \square \mathcal{N}_s^* \leq 0. \quad (2.25)$$

From the above derivative bounds, we have the following comparison result between two different space-time points, which was not quite possible in Hein-Naber's argument (ref. **Theorem 1.36** – time has to be equal):

**Corollary 2.17.** *If  $R(\cdot, t^*) \geq R_{\min}$ ,  $s < t^* \leq t_1, t_2$ , and  $s, t^*, t_1, t_2 \in I$ , then for  $x_1, x_2 \in M$  we have*

$$\mathcal{N}_s^*(x_1, t_1) - \mathcal{N}_s^*(x_2, t_2) \leq \left( \frac{n}{2(t^* - s)} - R_{\min} \right)^{\frac{1}{2}} d_{W_1}^{g_{t^*}}(\nu_{x_1, t_1}(t^*), \nu_{x_2, t_2}(t^*)) + \frac{n}{2} \log \left( \frac{t_2 - s}{t^* - s} \right). \quad (2.26)$$

**Remark 2.18.** *After we have introduced  $P^*$ -neighborhoods, we can view **Corollary 2.17** as a bound on the oscillation of  $\mathcal{N}_s^*$  over a  $P^*$ -neighborhood  $P := P^*(x, t; A, -T^-, T^+)$ :*

$$\text{osc}_P \mathcal{N}_s^* \leq 2 \left( \frac{n}{2(t-s-T^-)} - R_{\min} \right)^{1/2} A + \frac{n}{2} \log \left( \frac{t-s+T^+}{t-s-T^-} \right). \quad (2.27)$$

We first prove two bounds that will be useful for the proof of **Theorem 2.16** (which are also very interesting on their very own).

**Proposition 2.19.** *If  $d\nu = (4\pi\tau)^{-n/2} e^{-f} dg$  denotes the conjugate heat kernel based at  $(x_0, t_0)$  and  $R(\cdot, t_0 - \tau) \geq R_{\min}$ , then*

$$\int_M \tau (|\nabla f|^2 + R) d\nu_{t_0 - \tau} \leq \frac{n}{2} \quad (2.28)$$

$$\int_M \left( f - \mathcal{N}_{x_0, t_0} - \frac{n}{2} \right)^2 d\nu_{t_0 - \tau} \leq n - 2R_{\min}\tau. \quad (2.29)$$

*Proof.* Follows from direct computation and **Proposition 2.15**.  $\square$

*Proof of Theorem 2.16.* After a shift in time, we may assume WLOG that  $s = 0$ .

We first express the pointed Nash entropy in terms of the heat kernel. Consider a point  $(x, t) \in M \times I$  write  $K(x, t; y, 0) = (4\pi t)^{-n/2} e^{-f(y, 0)}$  and  $d\nu_0 = (4\pi t)^{-n/2} e^{-f(\cdot, 0)} dg$ . Then

$$\begin{aligned}\mathcal{N}_0^*(x, t) &= \int_M f(y, 0) K(x, t; y, 0) dg_0(y) - \frac{n}{2} \\ &= - \int_M K(x, t; y, 0) \log K(x, t; y, 0) dg_0(y) - \frac{n}{2} \log(4\pi t) - \frac{n}{2}.\end{aligned}$$

For any vector  $v \in T_x M$  with  $|v|_t = 1$ , we have

$$\int_M \partial_v K(x, t; y, 0) dg_0(y) = \partial_v \int_M K(x, t; y, 0) dg_0(y) = \partial_v 1 = 0.$$

Thus

$$\begin{aligned}\partial_v \mathcal{N}_0^*(x, t) &= - \int_M (\partial_v K(x, t; y, 0) \log K(x, t; y, 0) + \partial_v K(x, t; y, 0)) dg_0(y) \\ &= \int_M \partial_v K(x, t; y, 0) \left( f(y, 0) + \frac{n}{2} \log(4\pi t) - 1 \right) dg_0(y) \\ &= \int_M \partial_v K(x, t; y, 0) \left( f(y, 0) - \mathcal{N}_0^*(x, t) - \frac{n}{2} \right) dg_0(y) \\ &= \int_M \frac{\partial_v K(x, t; y, 0)}{K(x, t; y, 0)} \left( f(y, 0) - \mathcal{N}_0^*(x, t) - \frac{n}{2} \right) d\nu_0(y) \\ &\leq \left( \int_M \left( \frac{\partial_v K(x, t; y, 0)}{K(x, t; y, 0)} \right)^2 d\nu_0 \right)^{\frac{1}{2}} \left( \int_M \left( f - \mathcal{N}_0^*(x, t) - \frac{n}{2} \right)^2 d\nu_0 \right)^{\frac{1}{2}}.\end{aligned}$$

Applying **Theorem 2.11** and **Proposition 2.19** we obtain the bound

$$|\nabla \mathcal{N}_0^*|^2(x, t) \leq \frac{1}{2t} (n - 2R_{\min} t).$$

To obtain the second bound, we compute  $\square \mathcal{N}_0^*(x, t)$  as below:

$$\begin{aligned}\square \mathcal{N}_0^*(x, t) &= - \int_M \square_x (K(x, t; y, 0) \log K(x, t; y, 0)) dg_0(y) - \frac{n}{2t} \\ &= - \int_M (\partial_t K(x, t; y, 0) \log K(x, t; y, 0) + \partial_t K(x, t; y, 0)) \\ &\quad - \Delta_x K(x, t; y, 0) \log K(x, t; y, 0) - \Delta_x K(x, t; y, 0) - \frac{|\nabla_x K(x, t; y, 0)|^2}{K(x, t; y, 0)} dg_0(y) - \frac{n}{2t} \\ &= \int_M \left( \frac{|\nabla_x K(x, t; y, 0)|^2}{K(x, t; y, 0)} \right) d\nu_0 - \frac{n}{2t}.\end{aligned}$$

Then the desired bounds follow by **Theorem 2.11**.  $\square$

*Proof of Corollary 2.17.* WLOG assume  $s = 0$ . Denote by  $d\nu^i = (4\pi\tau_i)^{-n/2} e^{-f_i} dg$  the conjugate

heat kernels based at  $(x_i, t_i)$ ,  $i = 1, 2$ . By **Theorem 2.16** we have

$$\mathcal{N}_0^*(x_i, t_i) \leq \int_M \mathcal{N}_0^*(\cdot, t^*) d\nu_{t^*}^i \leq \mathcal{N}_0^*(x_i, t_i) + \frac{n}{2} \int_{t^*}^{t_i} \frac{dt}{t} = \mathcal{N}_0^*(x_i, t_i) + \frac{n}{2} \log \left( \frac{t_i}{t^*} \right). \quad (2.30)$$

Furthermore, due to the gradient bound (2.25) we have

$$\left| \int_M \mathcal{N}_0^*(\cdot, t) d\nu_{t^*}^1 - \int_M \mathcal{N}_0^*(\cdot, t^*) d\nu_{t^*}^2 \right| \leq \left( \frac{n}{2t^*} - 2R_{min} \right)^{1/2} d_{W_1}^{g_{t^*}}(\nu_{t^*}^1, \nu_{t^*}^2). \quad (2.31)$$

Combining the above bounds, we obtain the desired bound.  $\square$

### 2.3 Volume bounds on distance balls

Using the gradient bounds of the heat kernel and Nash entropy provided in previous subsection, here we prove certain volume bounds on distance balls. One of the main results in this subsection implies that, for spaces with lower Ricci curvature bounds,  $\mathcal{N}_{x_0, t_0}(\tau)$  is comparable to

$$V(x_0, r) := \log(r^{-n} |B(x, r)|).$$

**Theorem 2.20** (Lower volume bound on a distance ball). *If  $R \leq r^{-2}$  on  $B(x, t, r)$  and  $[t - r^2, t] \subset I$ , then*

$$|B(x, t, r)|_t \geq c \exp(\mathcal{N}_{x, t}(r^2)) r^n. \quad (2.32)$$

*Proof.* A point-picking argument similar to that of **Theorem 1.17**, involving the gradient bound of the Nash entropy in **Theorem 2.16**.  $\square$

Bamler was also able to provide another lower bound on the distance ball centered at the  $H_n$ -center, in absence of scalar curvature bound.

**Theorem 2.21.** *Suppose that  $[t - r^2, t] \subset I$  and that  $(z, t - r^2) \in M \times I$  is an  $H_n$ -center of some point  $(x, t) \in M \times I$ . If  $R(\cdot, t - r^2) \geq R_{min}$ , then we have*

$$\left| B(z, t - r^2, \sqrt{2H_n}r) \right|_{t-r^2} \geq c(R_{min}r^2) \exp(\mathcal{N}_{x, t}(r^2)) r^n, \quad (2.33)$$

where  $c(R_{min}r^2) = c \exp(-2(n - 2R_{min}r^2)^{\frac{1}{2}})$ .

*Proof.* After parabolic rescaling and application of a time-shift, we may assume that  $r = t = 1$ . Denote by  $d\nu = (4\pi\tau)^{-n/2} e^{-f} dg$  the measure of the conjugate heat kernel based at  $(x, 0)$  and set  $B := B(z, 0, 2\sqrt{H_n})$ . Then from (2.15)

$$\nu_0(B) \geq \frac{1}{2}. \quad (2.34)$$

By **Proposition 2.19** we have

$$\int_M \left| f - \mathcal{N}_{x, 1}(1) - \frac{n}{2} \right| d\nu_0 \leq \left( \int_M \left( f - \mathcal{N}_{x, 1} - \frac{n}{2} \right) d\nu_0 \right)^{1/2} \leq (n - 2R_{min})^{1/2}.$$

Therefore

$$\frac{1}{\nu_0(B)} \int_M f d\nu_0 \geq \mathcal{N}_{x,1}(1) + \frac{n}{2} - \frac{1}{\nu_0(B)} \int_B \left| f - \mathcal{N}_{x,1}(1) - \frac{n}{2} \right| d\nu_0 \geq \mathcal{N}_{x,1}(1) + \frac{n}{2} - 2(n - 2R_{\min})^{1/2}.$$

Denote  $u := (4\pi)^{n/2} e^{-f} / \nu_0(B)$ . It follows using (2.34) that  $\int_B u dg_0 = 1$  and

$$\begin{aligned} \int_B (\log u) u dg_0 &= -\frac{1}{\nu_0(B)} \int_B f d\nu_0 - \log(\nu_0(B)) + \frac{n}{2} \log(4\pi) \\ &\leq -\mathcal{N}_{x,1}(1) - \frac{n}{2} + 2(n - 2R_{\min})^{1/2} + \log 2 + \frac{n}{2} \log(4\pi). \end{aligned}$$

Applying Jensen's inequality to the convex function  $x \mapsto x \log x$  (just as in **Theorem 1.17**):

$$\log \left( \frac{1}{|B|_0} \int_B u dg_0 \right) \frac{1}{|B|_0} \int_B u dg_0 \leq \frac{1}{|B|_0} \int_B (\log u) u dg_0,$$

implying

$$-\log |B|_0 \leq \int_M (\log u) u dg_0 \leq -\mathcal{N}_{x,1}(1) + C + 2(n - 2R_{\min})^{1/2},$$

completing the proof of the theorem.  $\square$

**Remark 2.22.** *The significance of the convex function  $x \mapsto x \log x$  in the proofs of non-collapsing theorems (Perelman, Hein-Naber, then Bamler) is evident.*

**Theorem 2.23** (Upper volume bound on a distance ball). *If  $[t - r^2, t] \subset I$  and  $R \geq R_{\min}$  on  $M \times [t - r^2, t]$ , then for any  $1 \leq A < \infty$*

$$|B(x, t, Ar)|_t \geq C(R_{\min} r^2) \cdot \exp(\mathcal{N}_{x,t}(r^2)) \exp(C_0(n) A^2) r^n \leq C(R_{\min} r^2) \cdot \exp(C_0(n) A^2) r^n. \quad (2.35)$$

*Proof.* After a suitable parabolic rescaling and a time-shift, we may assume that  $t = 0$  and  $r = 2$ . Since by **Proposition 2.15** we have  $\mathcal{N}_{x,t}(1) \leq \mathcal{N}_{x,t}(4) + C(R_{\min})$ , it suffices to prove in the following that for some dimensional constant  $C < \infty$ ,

$$|B(x, t, A)|_0 \leq C \exp(\mathcal{N}_{x,t}(1)) \exp(CA^2). \quad (2.36)$$

Applying the maximum principle on the evolution equation for the scalar curvature yields  $R \geq -n/2$  on  $M \times [-1, 0]$ . By **Theorem 2.16** we have

$$-\mathcal{N}_{x,0}(1) \leq -\mathcal{N}_{x,0}(1) + CA \quad \text{on } B(x, 0, A). \quad (2.37)$$

Next, recall that we have

$$\mathcal{N}_{x,0}(1) = - \int_M (\log K(x, 0; \cdot, -1)) K(x, 0; \cdot, -1) dg_{-1} - \frac{n}{2} - \frac{n}{2} \log(4\pi).$$

Then we can find a point  $y \in M$  with

$$\log K(x, 0; y, -1) \geq -\mathcal{N}_{x,0}(1) - \frac{n}{2} - \frac{n}{2} \log(4\pi),$$

implying

$$K(x, 0; y, -1) \geq c \exp(-\mathcal{N}_{x,0}(1)). \quad (2.38)$$

Set  $u := K(\cdot, 0; y, -1)$ . By **Theorem 2.13** there exists a constant  $C_0 < \infty$  such that

$$\frac{|\nabla u|}{u} \leq C \sqrt{\log \left( \frac{C_0 \exp(-\mathcal{N}_{\cdot,0}(1))}{u} \right)}. \quad (2.39)$$

So by (2.37) there is a constant  $C_1 < \infty$  such that on  $B(x, 0, A)$ ,

$$\frac{|\nabla u|}{u} \leq C \sqrt{\log \left( \frac{C_1 \exp(-\mathcal{N}_{\cdot,0}(1)) + C_1 A}{u} \right)}. \quad (2.40)$$

So if we set

$$v := \sqrt{\log \left( \frac{C_1 \exp(-\mathcal{N}_{\cdot,0}(1)) + C_1 A}{u} \right)}$$

then by (2.39) and (2.40) we have

$$|\nabla v| \leq C \quad \text{on } B(x, 0, A), \quad v(x) \leq C\sqrt{A}.$$

This implies that  $v \leq C\sqrt{A} + CA \leq CA$  on  $B(x, 0, A)$ , so

$$u \geq c \exp(-CA^2) \exp(-\mathcal{N}_{x,0}(1)) \quad \text{on } B(x, 0, A). \quad (2.41)$$

Notice that since

$$\frac{d}{dt} \int_M K(\cdot, t; y, -1) dg_t = - \int_M RK(\cdot, t; y, -1) dg_t \leq C \int_M K(\cdot, t; y, -1) dg_t,$$

we have using (2.41)

$$c \exp(-CA^2) \exp(-\mathcal{N}_{x,0}(1)) |B(x, 0, A)|_0 \leq \int_M u dg_0 \leq C,$$

completing the proof of the theorem.  $\square$

## 2.4 Geometry of $P^*$ -parabolic neighborhoods

Now we are ready to introduce the  $P^*$ -parabolic neighborhoods (or in short,  $P^*$ -neighborhoods), which are of great importance in Bamler's proof of his improved  $\epsilon$ -regularity theorem. We shall begin with its definition.

**Definition 2.24** ( $P^*$ -parabolic neighborhood). *Suppose that  $(x_0, t_0) \in M \times I$ ,  $A, T^+, T^- \geq 0$  and  $t_0 - T^- \in I$ , then the  $P^*$ -**parabolic neighborhood**  $P^*(x_0, t_0; A, -T^-, T^+) \subset M \times I$  is defined as the set of points  $(x, t) \in M \times I$  with  $t_0 \in [t_0 - T^-, t_0 + T^+]$  and*

$$d_{W_1}^{g_{t_0-T^-}}(\nu_{x_0, t_0; t_0-T^-}, \nu_{x, t; t_0-T^-}) < A.$$

**Definition 2.25** (Forward/backward  $P^*$ -parabolic ball). *If  $(x_0, t_0) \in M \times I$  and  $r > 0$  such that  $t_0 - r^2 \in I$ , then the  $P^*$ -parabolic  $r$ -ball is defined as follows:*

$$P^*(x_0, t_0; r) := P^*(x_0, t_0; r, -r^2, r^2).$$

Similarly, we define the **forward and backward  $P^*$ -parabolic  $r$ -balls** by

$$P^{*+}(x_0, t_0; r) = P^*(x_0, t_0; r, 0, r^2), \quad P^{*-}(x_0, t_0; r) = P^*(x_0, t_0; r, -r^2, 0).$$

Recall the definition of the usual parabolic neighborhood: (we may omit  $-T^-$  or  $T^+$  if it is 0)

$$P(x_0, t_0; A, -T^-, T^+) := B(x_0, t_0, A) \times ([t_0 - T^-, t_0 + T^+] \cap I).$$

Recall the proof of **Theorem 1.38** and **Proposition 1.39**: after applying the point-picking argument, we find a (pointed) limit flow which exists in a fixed parabolic neighborhood of a point and then derive contradiction using the Nash entropy from there. We want to repeat the same procedure in the proof of Bamler's  $\epsilon$ -regularity theorem.  $P^*$ -parabolic neighborhoods, from its definition, is a collection of open sets which describe the behavior of the pointed Nash entropy near an appropriately chosen space-time better than the usual parabolic neighborhoods. Nevertheless, to actually prove our desired  $\epsilon$ -regularity theorem applying the argument of Hein-Naber's, ultimately we need to know something about the parabolic neighborhood in a pointed limit flow. Therefore, it is desirable to check if there is some type of containment relationship between the usual parabolic neighborhoods and  $P^*$ -parabolic neighborhoods. It turns out there are such relationships; before that, we have to check the containment relationships between different  $P^*$ -parabolic neighborhoods.

**Proposition 2.26** (Containment relationships between  $P^*$ -neighborhoods). *The following holds for any  $(x_1, y_1), (x_2, y_2) \in M \times I$  as long as the corresponding  $P^*$ -neighborhoods or balls are defined:*

1. *For any  $A \geq 0$  we have*

$$P^*(x_1, t_1; A, 0, 0) = B(x_1, t_1, A) \times \{t_1\}.$$

2. *If  $0 \leq A_1 \leq A_2$ ,  $0 \leq T_1^\pm \leq T_2^\pm$ , then*

$$P^*(x_1, t_1; A_1, -T_1^-, T_1^+) \subset P^*(x_1, t_1; A_2, -T_2^-, T_2^+).$$

3. *If  $A, T^\pm \geq 0$ , and  $(x_1, t_1) \in P^*(x_2, t_2; A, -T^-, T^+)$ , then*

$$(x_2, t_2) \in P^*(x_1, t_1; A, -(T^-, T^+), T^-).$$

and

$$P^*(x_2, t_2; A, -T^-, T^+) \subset P^*(x_1, t_1; 2A, -(T^- + T^+), T^-, T^+).$$

Likewise, if  $r > 0$  and  $(x_1, t_1) \in P^*(x_2, t_2; r)$ , then

$$(x_2, t_2) \in P^*(x_1, t_1; \sqrt{2}r) \text{ and } P^*(x_2, t_2; r) \subset P^*(x_1, t_1; 2r).$$

4. If  $A_1, A_2, T_1^\pm, T_2^\pm \geq 0$  and  $(x_1, t_1) \in P^*(x_2, t_2; A_2, -T^-, T_2^+)$ , then

$$P^*(x_1, t_1; A_1, -T^-, T^+) \subset P^*(x_2, t_2; A_1 + A_2, -(T_1^- + T_2^-), T_1^+ + T_2^+).$$

Likewise, if  $r_1, r_2 > 0$  and  $(x_1, t_1) \in P^*(x_2, t_2; r_2)$  then

$$P^*(x_1, t_1; r_1) \subset P^*(x_2, t_2; r_1 + r_2).$$

The same containment relationship also holds for the forward or backward parabolic balls, if  $t_1 \geq t_2$  or  $t_1 \leq t_2$ , respectively.

5. If  $r_1, r_2 > 0$  and  $P^*(x_1, t_1; r_1) \cap P^*(x_2, t_2; r_2) \neq \emptyset$ , then  $P^*(x_1, t_1; r_1) \subset P^*(x_2, t_2; 2r_1 + r_2)$ . Again, the same containment relationship also holds for the forward or backward parabolic balls, if  $t_1 \geq t_2$  or  $t_1 \leq t_2$ , respectively.

*Proof.* Follows from the definition of  $P^*$ -neighborhoods/balls + direct computation.  $\square$

Then we have the following comparison between the usual parabolic neighborhoods and  $P^*$ -neighborhoods.

**Proposition 2.27** (Containment relationships between  $P$ - and  $P^*$ -neighborhoods). *For any  $0 < \alpha \leq A < \infty$ ,  $K, T^\pm \geq 0$  the following holds if  $A' \geq \underline{A}'(\alpha, A, K, T^\pm)$ . Consider a point  $(x_0, t_0) \in M \times I$  and a scale  $r > 0$  with  $[t_0 - T^-r^2, t_0] \subset I$ . Then:*

1. If  $|Ric| \leq Kr^{-2}$  on  $P(x_0, t_0; Ar, -T^-r^2, T^+r^2)$  or on  $P^*(x_0, t_0; A'r, -T^-r^2, T^+r^2)$ , then

$$P(x_0, t_0; Ar, -T^-r^2, T^+r^2) \subset P^*(x_0, t_0; A'r, -T^-r^2, T^+r^2).$$

2. If  $|Ric| \leq Kr^{-2}$  on  $P(x_0, t_0; A'r, -T^-r^2, T^+r^2)$ , then

$$P^*(x_0, t_0; Ar, -T^-r^2, T^+r^2) \subset P(x_0, t_0; A'r, -T^-r^2, T^+r^2).$$

Above containment relationship is based on the following important proposition:

**Proposition 2.28.** *Assume that  $\alpha > 0$  and  $K < \infty$  and consider a point  $(x_0, t_0) \in M \times I$  and a scale  $r > 0$  with  $[t_0 - r^2, t_0] \subset I$ . Suppose that  $|Ric| \leq Kr^{-2}$  on  $P(x_0, t_0; \alpha r, -r^2)$ . Then*

$$d_{W_1}^{g_{t_0-r^2}}(\nu_{x_0, t_0; t_0-r^2}, \delta_{x_0}) \leq C(\alpha, K)r.$$

*Proof.* After parabolic rescaling and application of a time-shift, we may assume that  $r = 1$  and  $t_0 - r^2 = 0$ . Consider the function  $u \in C_c^0(P(x_0, 1; \alpha, -1))$  and the constants  $c_0(\alpha, K) \geq c_1(\alpha, K) > 0$  from the below lemma:

**Lemma 2.29.** *In the setting of **Proposition 2.28**, we can find a compactly supported function  $u \in C_c^0(P(x_0, t_0; \alpha r, -r^2))$  such that the following is true:*

1.  $0 \leq u \leq 1$ .
2.  $\square u \leq 0$  in the barrier and viscosity sense.
3. There are constants  $c_0(\alpha, K) \geq c_1(\alpha, K) > 0$  such that for all  $t \in [t_0 - r^2, t_0]$  we have  $\text{supp } u(\cdot, t) \subset B(x_0, t, c_0 r) \subset P(x_0, t_0; \alpha r, -r^2)$  and  $u \geq c_1$  on  $B(x_0, t, c_1 r)$ .

Then we obtain that

$$\nu_{x_0, 1; 0}(B(x_0, 0, c_0)) \geq \int_M u d\nu_{x_0, 1}(0) \geq u(x_0, 1) \geq c_1 \quad (2.42)$$

where the second to last “ $\geq$ ” follows from the property of conjugate heat kernel measure

$$\frac{d}{dt} \int_M u d\nu_{x_0, 1; t} = \int_M \square u d\nu_{x_0, 1; t} \leq 0. \quad (2.43)$$

Combining this with the bound (2.15), we have: for any  $H_n$ -center  $(z, 0)$  of  $(x_0, 1)$

$$d_0(z, x_0) \leq C(\alpha, K)$$

and therefore,

$$d_{W_1}^{g_0}(\nu_{x_0, 1; 0}, \delta_{x_0}) \leq d_{W_1}^{g_0}(\nu_{x_0, 1; 0}, \delta_z) + d_{W_1}^{g_0}(\delta_z, \delta_{x_0}) \leq \sqrt{H_n} + C(\alpha, K) \leq C(\alpha, K) \quad (2.44)$$

as desired.  $\square$

*Proof of Lemma 2.29.* After parabolic rescaling and application of a time-shift, we may assume that  $r = 1$  and  $t_0 - r^2 = 0$ . Let  $A < \infty$  be a constant whose value we will determine later. Choose a bump function  $\phi : [0, \infty) \rightarrow [0, 1]$  such that  $\phi \equiv 1$  on  $[0, \frac{1}{3}\alpha e^{-K}]$  and  $\phi \equiv 0$  on  $[\frac{2}{3}\alpha e^{-K}, \infty)$  and such that on  $[0, \infty)$  we have

$$\phi' \leq 0, \quad A\phi' + \phi'' \geq -C_0(A, \alpha, K)\phi.$$

Define

$$u(x, t) := e^{-C_0(A, \alpha, K)t} \phi(d_t(x_0, x)).$$

By Laplace comparison we have for any  $t \in [0, 1]$  in the barrier sense

$$\begin{aligned} \square u &= \left( \frac{d}{dt} - \Delta \right) u \leq e^{-C_0(A, \alpha, K)t} (\phi' \partial_t d_t(x_0, \cdot) - C_0(A, \alpha, K)\phi - \phi' \Delta_{g_t} d_t(x_0, \cdot) - \phi'') \\ &\leq e^{-C_0(A, \alpha, K)t} \left( -K\phi' d_t(x_0, \cdot) - \phi' \frac{(n-1)\sqrt{\frac{K}{n-1}}}{\tanh\left(\sqrt{\frac{K}{n-1}} d_t(x_0, \cdot)\right)} - \phi'' - C_0(A, \alpha, K)\phi \right) \\ &\leq e^{-C_0(A, \alpha, K)t} \left( - \left( K\alpha e^{-K} + \frac{(n-1)\sqrt{\frac{K}{n-1}}}{\tanh\left(\frac{1}{3}\alpha e^{-K} \sqrt{\frac{K}{n-1}}\right)} \right) \phi' - \phi'' - C_0(A, \alpha, K)\phi \right). \end{aligned}$$

So if  $A = A(\alpha, K)$  is chosen to be larger than the term in parenthesis, then  $\square u \leq 0$  in the barrier and hence also in the viscosity sense.  $\square$

Now we are ready to prove **Proposition 2.27**.

*Proof of Proposition 2.27.*

- (a) Recall the standard distance distortion estimate in Ricci flow: given the Ricci curvature bound  $|\text{Ric}| \leq Kr^{-2}$  on  $P(x_0, t_0; Ar, -T^-r^2, T^+r^2)$ , for  $x, y \in B(x_0, t_0, Ar)$  and  $t, t' \in [t_0 - T^-r^2, t_0 + T^+r^2] \cap I$  we have

$$\exp(-Kr^{-2}(t' - t))d_t(x, y) \leq d_{t'}(x, y) \leq \exp(Kr^{-2}(t' - t))d_t(x, y). \quad (2.45)$$

Now suppose  $(y, t') \in P(x_0, t_0; Ar, -T^-r^2, T^+r^2)$ . By **Proposition 2.28** and **Lemma 2.2** (monotonicity of  $W_1$ -distance):

$$\begin{aligned} d_{W_1}^{g_{t_0 - T^-r^2}}(\nu_{x_0, t_0; t_0 - T^-r^2}, \nu_{y, t'; t_0 - T^-r^2}) &\leq d_{W_1}^{g_{t_0 - T^-r^2}}(\nu_{x_0, t_0; t_0 - T^-r^2}, \delta_{x_0}) + d_{W_1}^{g_{t_0 - T^-r^2}}(\delta_{x_0}, \nu_{y, t'; t_0 - T^-r^2}) \\ &\leq C(K, \alpha)r + d_{W_1}^{g_{t_0 - T^-r^2}}(\nu_{x_0, t_0 - T^-r^2; t_0 - T^-r^2}, \nu_{y, t'; t_0 - T^-r^2}) \\ &\leq C(K, \alpha)r + d_{t'}(x_0, y) \\ &\leq A'r \quad \text{by the distortion estimate above.} \end{aligned}$$

Hence we have the first inclusion for assertion (a). Now we show the second inclusion relationship: suppose we are given the bound  $|\text{Ric}| \leq Kr^{-2}$  on  $P^*(x_0, t_0; A'r, -T^-r^2, T^+r^2)$ . Choose  $r^* \in (0, r]$  maximal with the property that  $|\text{Ric}| \leq Kr^{*2}$  on  $P(x_0, t_0; Ar^{*2}, -T^-r^{*2}, T^+r^{*2})$ . If  $r^* = r$ , then we are done by the previous conclusion. If  $r^* < r$ , then by the previous conclusion

$$P(x_0, t_0; Ar^*, -T^-r^{*2}, T^+r^{*2}) \subset P^*(x_0, t_0; A'r^*, -T^-r^{*2}, T^+r^{*2}).$$

So  $|\text{Ric}| \leq Kr^{-2} < r^{*2}$  on  $P(x_0, t_0; Ar^*, -T^-r^{*2}, T^+r^{*2})$ , contradicting the maximal choice of  $r^*$ .

- (b) By parabolic rescaling, we may assume that  $r = 1$ . Furthermore, it is possible to reduce the problem to the case  $T^- = 0$ : by **Proposition 2.28**, for any  $(x, t) \in P^*(x_0, t_0; A, -T^-, T^+)$  we have

$$\begin{aligned} d_{W_1}^{g_{t_0 - T^-}}(\nu_{x_0, t_0 - T^-; t_0 - T^-}, \nu_{x, t; t_0 - T^-}) &\leq d_{W_1}^{g_{t_0 - T^-}}(\nu_{x_0, t_0 - T^-; t_0 - T^-}, \delta_{x_0}) + d_{W_1}^{g_{t_0 - T^-}}(\delta_{x_0}, \nu_{x, t; t_0 - T^-}) \\ &\leq C(\alpha, K, T^-) + A. \end{aligned}$$

Thereore

$$P^*(x_0, t_0; A, -T^-, T^+) \subset P^*(x_0, t_0 - T^-; C(\alpha, K, T^-) + A, 0, T^- + T^+).$$

By the distance distortion estimate (2.45), we also have

$$P(x_0, t_0; A', -T^-, T^+) \supset P(x_0, t_0 - T^-; c(K, T^-)A', 0, T^- + T^+).$$

So after replacing  $t_0$  with  $t_0 - T^-$ ,  $T^+$  with  $T^- + T^+$  and adjusting  $A'$  appropriately, we may indeed assume in the following that  $T^- = 0$  and write  $T = T^+$ . After a time-shift, we may furthermore assume that  $t_0 = 0$ . So we have reduced the problem to showing that if  $A' \geq \underline{A}(A, K, T)$  and  $|\text{Ric}| \leq K$  on  $P(x_0, 0; A', T)$ , then

$$P^*(x_0, 0; A, T) \subset P(x_0, 0; A', T'). \quad (2.46)$$

Fix  $\alpha, K, A, T$ . We will determine a lower bound of the form  $A' \geq \underline{A}(A, K, T)$  in the course of the proof. Choose  $T^* \in [0, T]$  maximal with the property that for all  $T' \in [0, T^*)$  we even have

$$P^*(x_0, 0; A, T') \subset P(x_0, 0; \frac{A'}{2}, T'). \quad (2.47)$$

Assuming  $A' > 2A$ , we must have  $T^* > 0$  due to the monotonicity of  $d_{W_1}$ . Our goal will be to show that we can improve (2.47). For this purpose, consider some  $(x, t) \in P^*(x_0, 0; A, T')$ . Then by (2.47) and a basic distance distortion estimate we have for  $A' \geq \underline{A}(K, A, T)$ ,

$$P(x, t; 1, -t) \subset P(x_0, 0; A', T').$$

Therefore by **Proposition 2.28** we have for  $A' \geq \underline{A}(K, A, T)$

$$d_0(x_0, x) = d_{W_1}^{g_0}(\delta_{x_0}, \delta_x) \leq d_{W_1}^{g_0}(\delta_{x_0}, \nu_{x, t; 0}) + d_{W_1}^{g_0}(\nu_{x, t; 0}, \delta_x) \leq A + C(K, T) \leq \frac{1}{4}A',$$

implying  $(x, t) \in P(x_0, 0; \frac{A'}{4}, T')$ . So by continuity and the maximal choice of  $T^*$ , we must have  $T^* = T$ , proving (2.46). □

Using **Proposition 2.27**, below we prove two important geometric properties of  $P^*$ -neighborhoods. Define the  **$P^*$ -time slice**

$$S_t := P^*(x_0, t_0; A, -T^-, T^+) \cap (M \times \{t\}). \quad (2.48)$$

**Theorem 2.30** (Geometry of time slices of  $P^*$ -neighborhoods). *Let  $\alpha > 0$ ,  $A, T^\pm \geq 0$  and consider a point  $(x_0, t_0) \in M \times I$  and a scale  $r > 0$  with  $[t_0 - (T^- + \alpha)r^2, t_0] \subset I$ . Then for any  $t \in [t_0 - T^-r^2, t_0 + T^+r^2]$  the time slice  $S_t$  defined above satisfies the following volume bound:*

$$|S_t|_t \leq C(A, T^-, T^+, \alpha) \exp(\mathcal{N}_{x_0, t_0}(T^-r^2))r^n. \quad (2.49)$$

Moreover, for any  $A' < \infty$  we have the following bound on the  $A'r$ -neighborhood of  $S_t$ :

$$|B(S_t, t, A'r)|_t \leq C(A, A', T^-, T^+, \alpha) \exp(\mathcal{N}_{x_0, t_0}(T^-r^2))r^n. \quad (2.50)$$

*Proof.* By parabolic rescaling and application of a time-shift, we may assume that  $r = 1$  and  $t_0 - T^- = 0$ . Let  $(z_0, 0)$  be an  $H_n$ -center of  $(x_0, t_0)$ . Fix some time  $t \in [t_0 - T^-, t_0 + T^+]$ ,

$x \in S_t$  and choose an  $H_n$ -center  $(z, 0)$  of  $(x, t)$ . Then

$$\begin{aligned} d_0(z_0, z) &= d_{W_1}^{g_0}(\delta_{z_0}, \delta_z) \\ &\leq d_{W_1}^{g_0}(\delta_{z_0}, \nu_{x_0, t_0; 0}) + d_{W_1}^{g_0}(\nu_{x_0, t_0; 0}, \nu_{x, t; 0}) + d_{W_1}^{g_0}(\nu_{x, t; 0}, \delta_z) \\ &\leq \sqrt{H_n t_0} + A + \sqrt{H_n t} \leq C(A, T^-, T^+). \end{aligned}$$

So by **Theorem 2.9** there is a constant  $D(A, T^-, T^+) < \infty$  such that for  $B := B(z_0, 0, D)$  we have

$$\nu_{x, t; 0}(B) \geq \nu_{x, t; 0}(B(z, 0, \sqrt{2H_n t})) \geq \frac{1}{2}. \quad (2.51)$$

Let now  $u \in C^0(M \times [0, t]) \cap C^\infty(M \times (0, t])$  be the solution to the heat equation  $\square u = 0$  with initial condition  $u(\cdot, 0) = \chi_B$ . By (2.51) we have  $u \geq \frac{1}{2}$  on  $S_t$  and since

$$\frac{d}{dt} \int_M u dg_t = - \int_M u R dg_t \leq \frac{n}{2\alpha} \int_M u dg_t \quad (2.52)$$

(by the evolution equation of  $R$  under the Ricci flow), we obtain

$$\frac{1}{2} |S_t|_t \leq \int_M u(\cdot, t) dg_t \leq \exp\left(\frac{nt}{2\alpha}\right) |B|_0. \quad (2.53)$$

Since  $d_{W_1}^{g_0}(\delta_{z_0}, \nu_{x_0, t_0}(0)) \leq \sqrt{H_n t_0}$ , we can use **Proposition 2.15** (properties of  $\mathcal{N}$ ) and **Corollary 2.17** to deduce that  $\mathcal{N}_{z_0, 0}\left(\frac{\alpha}{2}\right) \leq \mathcal{N}_{x_0, t_0}\left(\frac{\alpha}{2}\right) + C(T^-, \alpha)$ . So, using **Theorem 2.23** (upper volume bound on distance ball), we get

$$|B|_0 \leq C(A, T^-, T^+, \alpha) \exp\left(\mathcal{N}_{z_0, 0}\left(\frac{\alpha}{2}\right)\right) \leq C(A, T^-, T^+, \alpha) \exp\left(\mathcal{N}_{x_0, t_0}\left(\frac{\alpha}{2}\right)\right). \quad (2.54)$$

Combining (2.55) with (2.53) yields (2.49).

Lastly, we remark that (2.50) can be reduced to (2.49), because **Proposition 2.26** implies

$$B(S_t, t, A') \times \{t\} \subset P^*(x_0, t_0; A + A', -T^-, T^+). \quad (2.55)$$

□

An important application of **Theorem 2.30** is the following covering theorem:

**Theorem 2.31** (Covering theorem of  $P^*$ -balls). *Let  $\lambda_0 > 0$ ,  $A, T^\pm \geq 0$  and consider a point  $(x_0, t_0) \in M \times I$  and a scale  $r > 0$  with  $[t_0 - (T^- + \lambda_0^2)r^2, t_0] \subset I$ . Then for any subset  $X \subset P^*(x_0, t_0; Ar, -T^-r^2, T^+r^2)$  and  $\lambda \in (0, \lambda_0]$  we can find points  $(y_1, s_1), \dots, (y_N, s_N) \in X$  with the property that*

$$X \subset \bigcup_{i=1}^N P^*(y_i, s_i; \lambda r), \quad N \leq C(A, T^-, T^+, \lambda_0) \lambda^{-n-2}. \quad (2.56)$$

*Using the same argument, one also obtains a similar covering theorem for general  $P^*$ -neighborhoods.*

*Proof.* After parabolic rescaling, we may assume that  $r = 1$ . Next, we may assume that  $\lambda \in (0, \frac{\lambda_0}{3}]$  and aim to prove the theorem for  $\lambda$  replaced with  $3\lambda$ . To do this, choose a maximal collection of

points  $(y_1, s_1), \dots, (y_N, s_N) \in X$  with

$$P^*(y_i, s_i; \lambda) \cap P^*(y_j, s_j; \lambda) = \emptyset \quad \text{for all } i \neq j.$$

To see that these points satisfy the covering identity in (2.56) with  $\lambda$  replaced with  $3\lambda$ , consider a point  $(y', s') \in X$ . By maximality we must have  $P^*(y', s'; \lambda) \cap P^*(y_i, s_i, \lambda) \neq \emptyset$  for some  $i \in \{1, \dots, N\}$ , which implies using **Proposition 2.26** that  $(y', s') \in P^*(y', s'; \lambda) \subset P^*(y_i, s_i, 3\lambda)$ .

It remains to derive an upper bound on  $N$ . For this purpose, let  $\beta \in (0, 1/2]$  be some constant whose value we will determine later and observe that there is some time  $t^* \in [t_0 - T^- - 2\beta\lambda^2, t_0 + T^+ - \beta\lambda^2]$  and a subset  $\mathcal{I} \subset \{1, \dots, N\}$  with the property that

$$|\mathcal{I}| \geq \lfloor c(T^-, T^+, \beta)\lambda^2 N \rfloor \quad (2.57)$$

and  $t^* \in [s_i - 2\beta\lambda^2, s_i - \beta\lambda^2]$  for all  $i \in \mathcal{I}$ . For each  $i \in \mathcal{I}$  let  $(z_i, t^*)$  be an  $H_n$ -center of  $(y_i, s_i)$ . If  $\beta \leq \bar{\beta}$ , then for all  $i, j \in \mathcal{I}$ ,  $i \neq j$ ,

$$\begin{aligned} d_{t^*}(z_i, z_j) &\geq d_{W_1}^{g_{t^*}}(\nu_{y_i, s_i; t^*}, \nu_{y_j, s_j; t^*}) - d_{W_1}^{g_{t^*}}(\nu_{y_i, s_i; t^*}, \delta_{z_i}) - d_{W_1}^{g_{t^*}}(\delta_{z_j}, \nu_{y_j, s_j; t^*}) \\ &\geq d_{W_1}^{g_{s_i - \lambda^2}}(\nu_{y_i, s_i; s_i - \lambda^2}, \nu_{y_j, s_j; s_i - \lambda^2}) - 2\sqrt{2H_n\beta\lambda^2} \\ &\geq \lambda - 2\sqrt{2H_n\beta\lambda^2} \geq \frac{1}{2}\lambda. \end{aligned}$$

So the balls  $B_i := B(z_i, t^*, \frac{\lambda}{4})$ ,  $i \in \mathcal{I}$ , are pairwise disjoint. By **Theorem 2.21**, **Proposition 2.15**, **Corollary 2.17**, plus the evolution equation of scalar curvature under Ricci flow, we have, if  $\beta \leq \bar{\beta}$  is chosen such that  $\sqrt{2H_n} \cdot 2\beta \leq \frac{1}{4}$ ,

$$|B_i|_{t^*} \geq c(\lambda_0, \beta) \exp(\mathcal{N}_{y_i, s_i}(\lambda^2)) \lambda^n \geq c(A, T^-, T^+, \lambda_0, \beta) \exp(\mathcal{N}_{x_0, t_0}(\lambda_0^2/9)) \lambda^n. \quad (2.58)$$

On the other hand, **Theorem 2.30** implies that for  $S_{t^*}$  we have

$$|B(S_{t^*}, t^*, \lambda_0/4)|_{t^*} \leq C(A, T^-, T^+, \lambda_0) \exp(\mathcal{N}_{x_0, t_0}(\lambda_0^2/9)). \quad (2.59)$$

Combining (2.57)-(2.59), we obtain

$$\lfloor c(T^-, T^+, \beta)\lambda^2 N \rfloor \leq |\mathcal{I}| \leq C(A, T^-, T^+, \lambda_0, \beta)\lambda^{-n},$$

which finishes the proof.  $\square$

## 2.5 Bamler's improved $\epsilon$ -regularity theorem

**Theorem 2.32** (Bamler's improved  $\epsilon$ -regularity theorem). *There is a constant  $\epsilon(n) > 0$  such that the following holds. Consider a point  $(x, t) \in M \times I$ ,  $r > 0$  with  $[t - r^2, t] \subset I$ . If  $\mathcal{N}_{x, t}(r^2) \geq -\epsilon$ , then the regularity scale  $r_{Rm}(x, t) \geq \epsilon r$ .*

**Remark 2.33.** *Two major improvements compared to Hein-Naber's:*

1. absence of global scalar curvature and global  $(\mu-)$ entropy bound;

2. dependence on the pointed Nash entropy alone.

Similar to  $\epsilon$ -regularity theorems states previously, we would like to employ a point-picking argument to derive our desired conclusion. Prior to this step, we first prove the following proposition:

**Proposition 2.34.** *Let  $A > 0$  and assume that  $10Ar_{Rm}(x, t) \leq \frac{1}{2}$ . Then there is a point  $(x', t') \in P^{*-}(x, t; 10Ar_{Rm}(x, t))$  such that*

$$r_{Rm}(x', t') \leq r_{Rm}(x, t) \quad \text{and} \quad r_{Rm} \geq \frac{1}{10} r_{Rm}(x', t') \quad \text{on} \quad P^{*-}(x', t'; Ar_{Rm}(x', t')).$$

*Proof.* We can iteratively pick a maximal sequence  $(x'_0, t'_0) := (x, t), (x'_1, t'_1), \dots$  such that for  $r'_i := r_{Rm}(x'_i, t'_i)$  we have

$$(x'_{i+1}, t'_{i+1}) \in P^{*-}(x'_i, t'_i; Ar'_i), \quad r'_{i+1} < \frac{1}{10} r'_i.$$

Since  $\inf_{M \times [0,1]} r_{Rm} > 0$ , this sequence must be finite. We claim that its last element  $(x'_k, t'_k) =: (x', t')$  has the desired properties. For this purpose, we will show by induction that

$$(x', t') \in P^{*-}(x'_i, t'_i; 10Ar'_i). \tag{2.60}$$

The proposition follows for  $i = 0$ . (2.60) is trivially true for  $i = k$ . If it is true for  $i + 1$ , then by **Proposition 2.26**, we have

$$(x', t') \in P^{*-}(x'_{i+1}, t'_{i+1}; 10Ar'_{i+1}) \subset P^{*-}(x'_i, t'_i; A(r'_i + 10r'_{i+1})) \subset P^{*-}(x'_i, t'_i; 10Ar'_i),$$

which concludes the induction.  $\square$

*Proof of Theorem 2.32.* Without loss of generality, we may assume that  $r = 1$  and  $t = 1$ . We first apply a point-picking procedure. Consider a sequence of counterexamples  $(M_i, (g_{i,t})_{t \in [0,1]})$ ,  $(x_i, t_i = 1) \in M_i \times [0, 1]$  to the theorem for some sequence  $\epsilon_i \rightarrow 0$ . Then  $r_i^* := r_{Rm}(x_i, t_i) \rightarrow 0$ . Choose a sequence  $A_i \rightarrow \infty$  with  $10A_i r_i^* \rightarrow 0$ . Let  $(x'_i, t'_i) \in P^{*-}(x_i, t_i, 10A_i r_i^*)$  be the point from the above proposition and set  $r'_i := r_{Rm}(x'_i, t'_i) \rightarrow 0$ . Since  $10A_i r_i^* \rightarrow 0$  and  $\epsilon_i \rightarrow 0$ , we obtain from **Corollary 2.17** that for all  $\tau \in (0, t'_i - \frac{1}{2}]$

$$0 \geq \mathcal{N}_{x'_i, t'_i}(\tau) \geq \mathcal{N}_{x'_i, t'_i}(t'_i - \frac{1}{2}) = \mathcal{N}_{\frac{1}{2}}^*(x'_i, t'_i) \rightarrow 0.$$

Note that here we have used the fact that  $R \geq -n$  on  $M \times [\frac{1}{2}, 1]$ , following from the scalar curvature evolution equation.

By **Proposition 2.27**, we can find a sequence  $A'_i \leq A_i$ ,  $A'_i \rightarrow \infty$  such that

$$P^-(x'_i, t'_i; A'_i r'_i) \subset P^{*-}(x'_i, t'_i; A_i r'_i),$$

which implies that

$$r_{Rm} \geq \frac{1}{10} r'_i \quad \text{on} \quad P^-(x'_i, t'_i; A'_i r'_i).$$

Therefore, after applying a time-shift of  $-t'_i$  and parabolic rescaling by  $r_i'^{-2}$ , we obtain a sequence of pointed flows of the form  $(M'_i, (g'_{i,t})_{t \in [-(A'_i)^2, 0]}, x'_i)$  with  $r_{Rm} \geq \frac{1}{10}$  on  $P^-(x'_i, 0; A'_i)$  and  $\mathcal{N}_{x'_i, 0}(T) \rightarrow 0$

for any fixed  $T > 0$ .

By **Theorem 2.20**, the injectivity radius at  $(x'_i, 0)$  is uniformly bounded from below. So, after passing to a subsequence, these pointed flows converge to a smooth and complete pointed ancient flow of the form  $(M_\infty, (g_{\infty, t})_{t \leq 0}, x_\infty)$  with  $r_{Rm} \geq \frac{1}{10}$  everywhere. By **Proposition 2.28**, **Theorem 2.9** and **Theorem 2.12**, it follows moreover that the conjugate heat kernels  $K_i(x'_i, 0; \cdot, \cdot) =: (4\pi\tau)^{-n/2} e^{-f_i}$  on  $(M', (g'_t)_{t \in [-(A'_i)^2, 0]})$ , based at  $(x'_i, 0)$ , converge locally uniformly to a positive solution  $v_\infty = (4\pi\tau)^{-n/2} e^{-f_\infty} \in C^\infty(M_\infty \times \mathbb{R}_-)$  of the conjugate heat equation.

Since for any fixed  $T > 0$  we have  $\mathcal{N}_{x'_i, 0}(T) \rightarrow 0$ , we obtain using **Proposition 2.15** that for any fixed  $T > 0$ . By **Theorem 2.20**, the injectivity radius at  $(x'_i, 0)$  is uniformly bounded below. So, after passing to a subsequence, these pointed flows converge (in the Cheeger-Gromov-Hamilton sense) to a smooth and complete pointed ancient flow of the form  $(M_\infty, (g_{\infty, t})_{t \leq 0}, x_\infty)$  with  $r_{Rm} \geq \frac{1}{10}$  everywhere. By **Proposition 2.28**, **Theorem 2.9**, **Theorem 2.12**, it follows moreover that the conjugate heat kernels  $K_i(x'_i, 0; \cdot, \cdot) =: (4\pi\tau)^{-n/2} e^{-f_i} \in C^\infty(M_\infty \times \mathbb{R}_-)$  of the conjugate heat equation.

Since for any fixed  $T > 0$  we have  $\mathcal{N}_{x'_i, 0}(T) \rightarrow 0$ , we obtain using **Proposition 2.15** that for any fixed  $T > 0$

$$\mathcal{W}[g'_{i, -T}, f_{i, -T}, T] = \frac{d}{d\tau} (\tau \mathcal{N}_{x'_i, 0}(\tau)) \Big|_{\tau=T} \rightarrow 0,$$

implying

$$\int_{-T}^0 \int_M \tau \left| \text{Ric} + \nabla^2 f_{i, t} - \frac{1}{2\tau} g_{i, t} \right|^2 (4\pi\tau)^{-n/2} e^{-f_{i, t}} dg_{i, t} dt \rightarrow 0.$$

Passing this to the limit implies that on  $M_\infty \times \mathbb{R}_-$  we have  $\text{Ric} + \nabla^2 f_\infty - \frac{1}{2\tau} g_\infty = 0$ . So  $(M_\infty, (g_{\infty, t})_{t \leq 0}, x_\infty)$  must be a gradient shrinking soliton ( $f_\infty$  is the soliton potential). Due to the uniform curvature bound, this implies that the limit  $(M_\infty, (g_{\infty, t})_{t \leq 0})$  must be flat, due to the soliton symmetry. We therefore obtain a contradiction to  $r_{Rm}(x'_i, 0) = r'_i \rightarrow 0$ .  $\square$